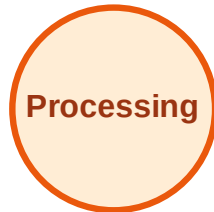


BFS Traversal

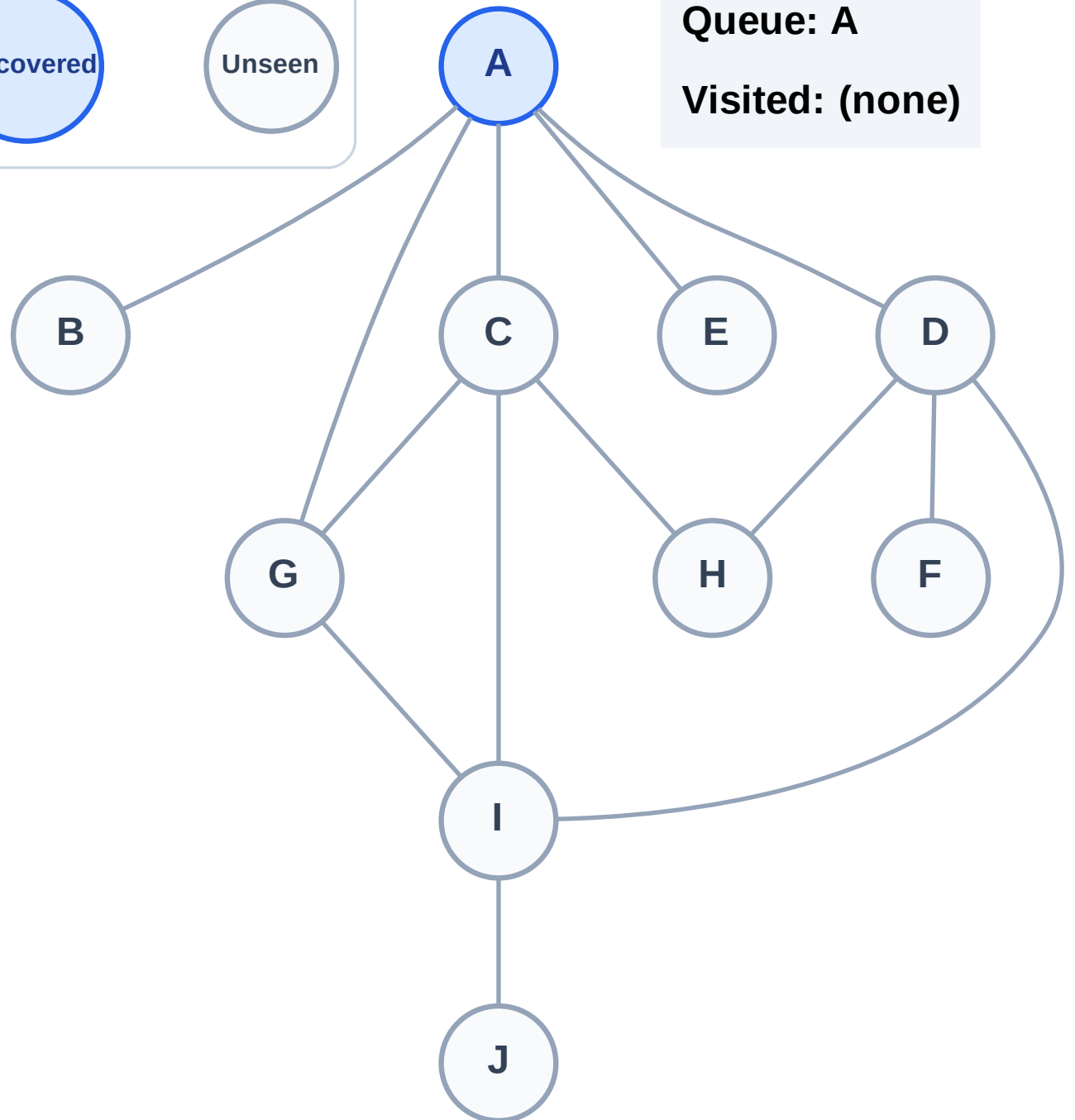
Step 1: Start at A

Legend



Queue: A

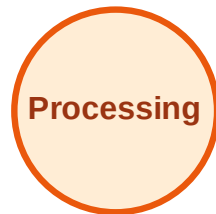
Visited: (none)



BFS Traversal

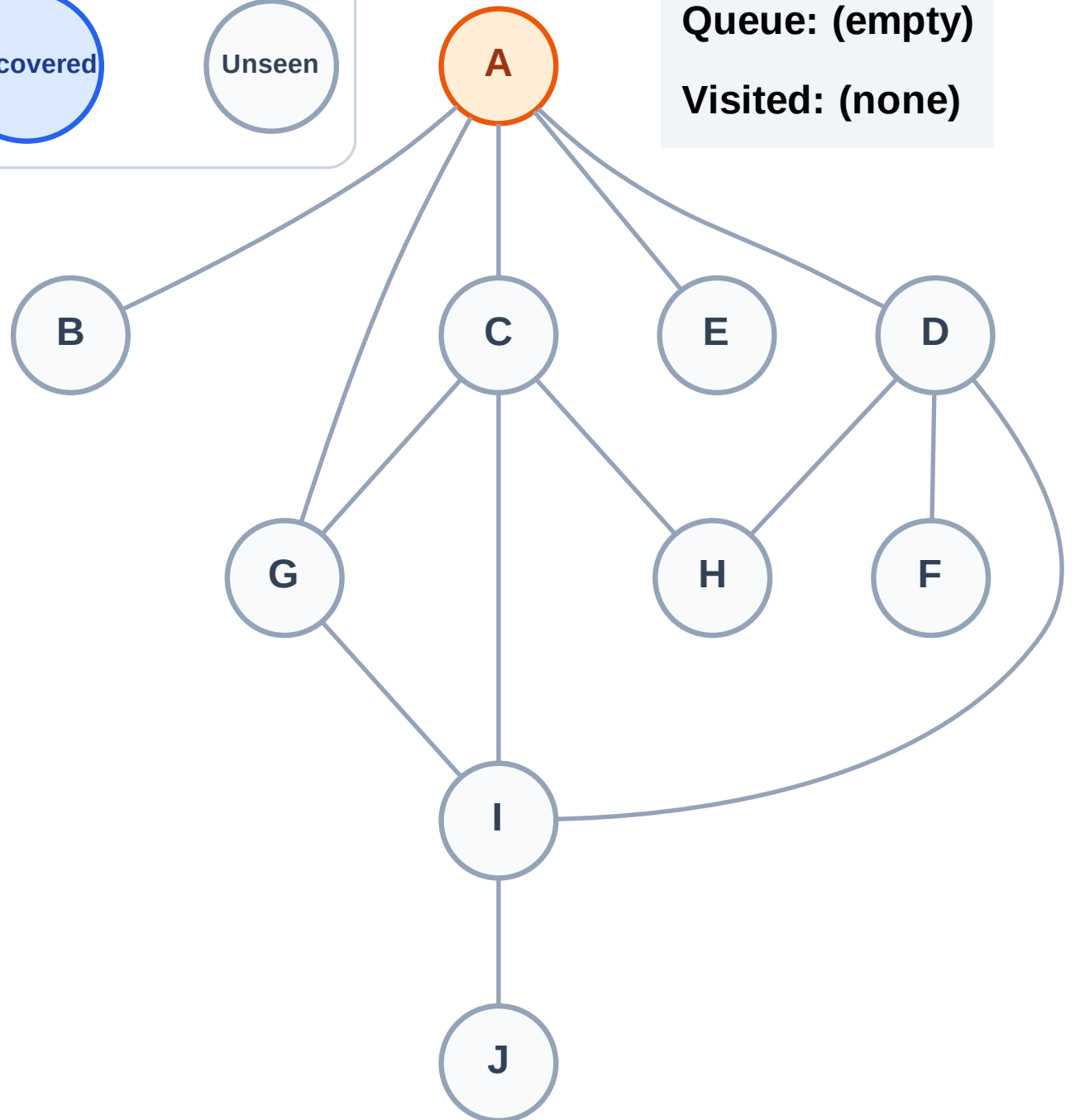
Step 2: Process node A

Legend



Queue: (empty)

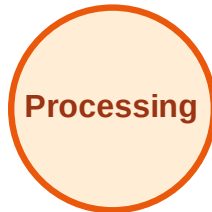
Visited: (none)



BFS Traversal

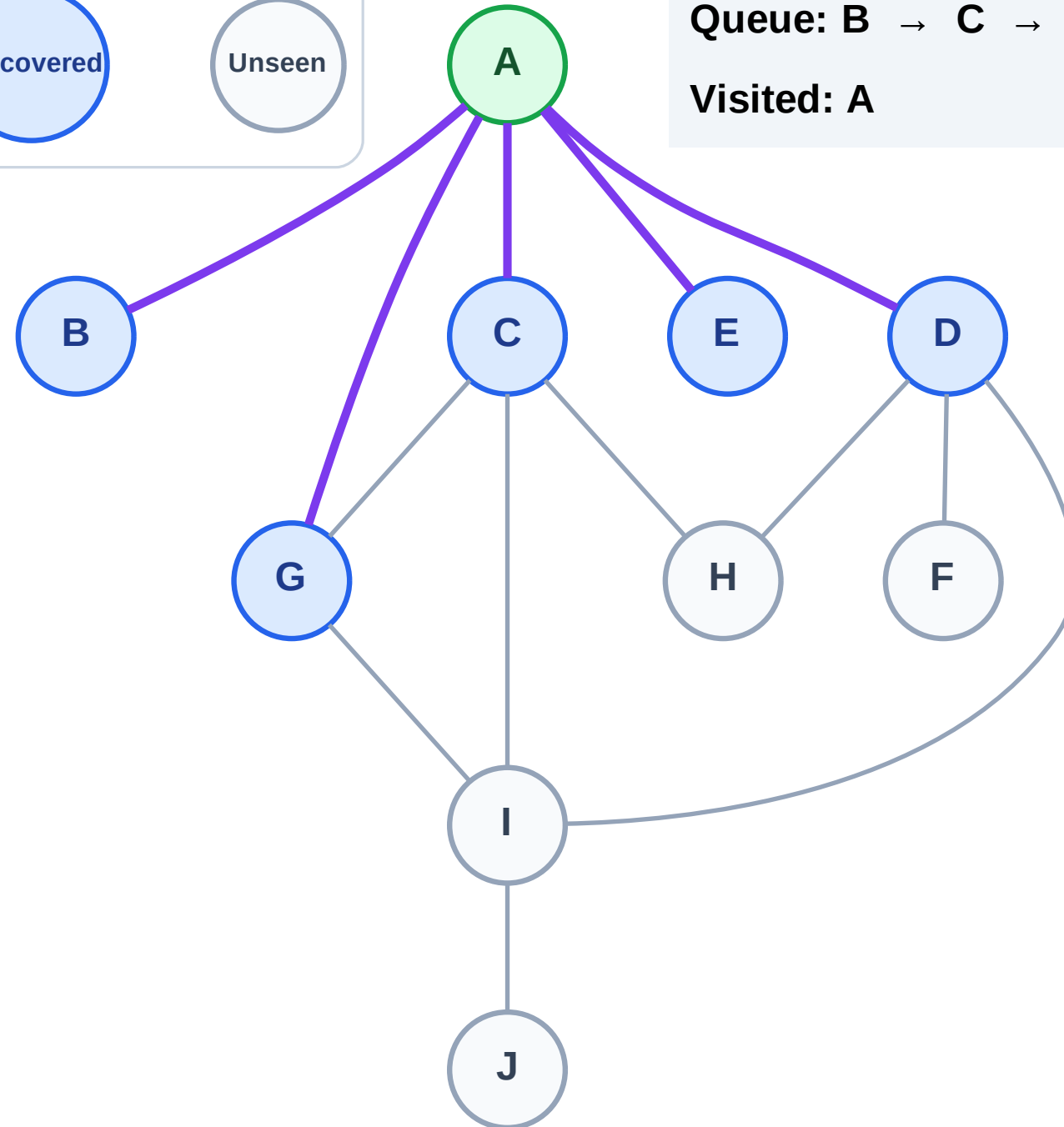
Step 3: Discover B, C, D, E, G from A

Legend



Queue: B → C → D → E → G

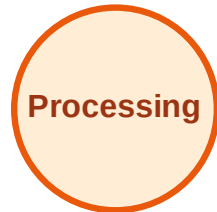
Visited: A



BFS Traversal

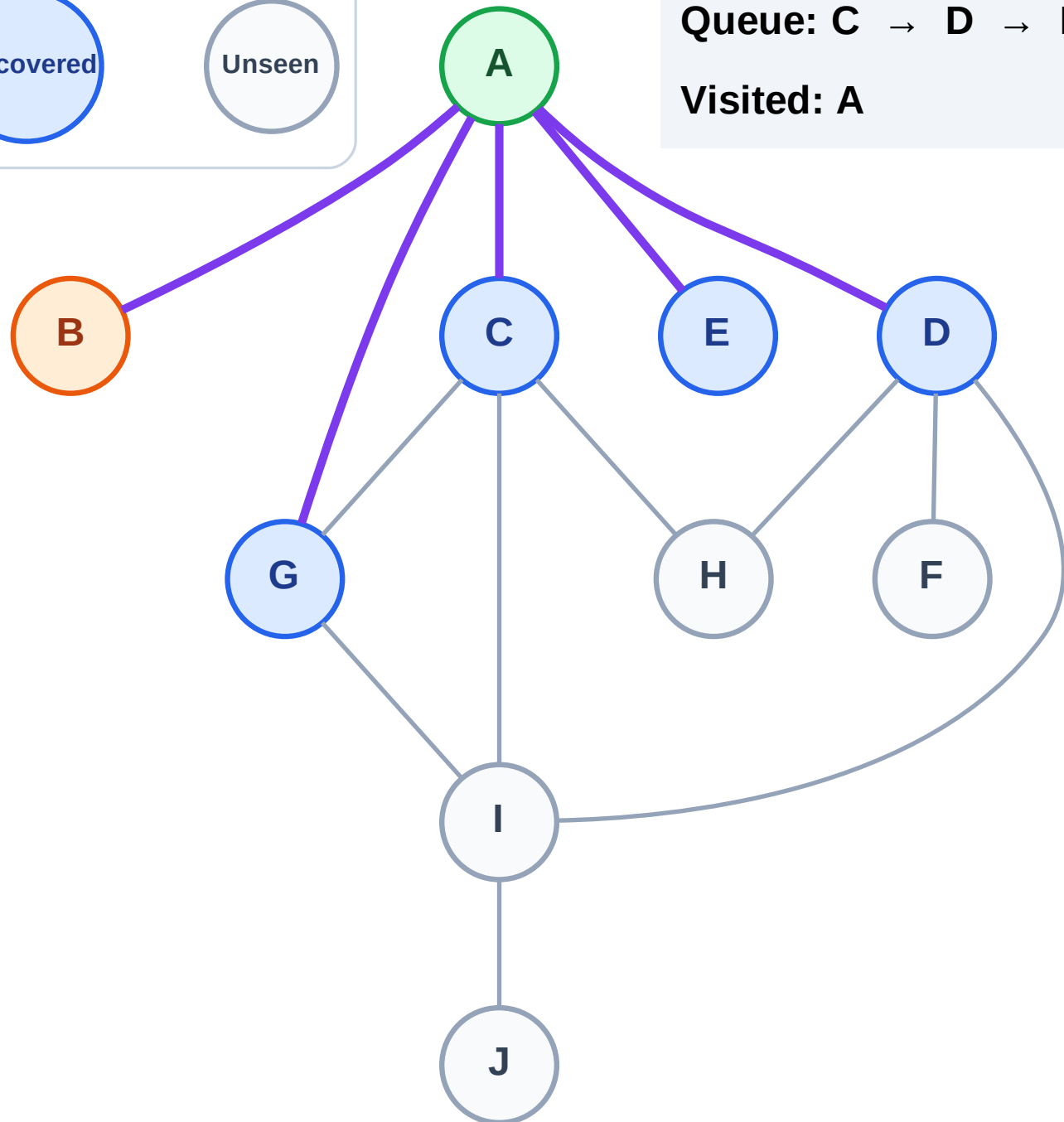
Step 4: Process node B

Legend



Queue: C → D → E → G

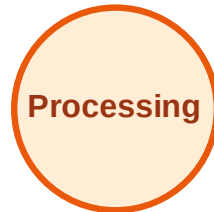
Visited: A



BFS Traversal

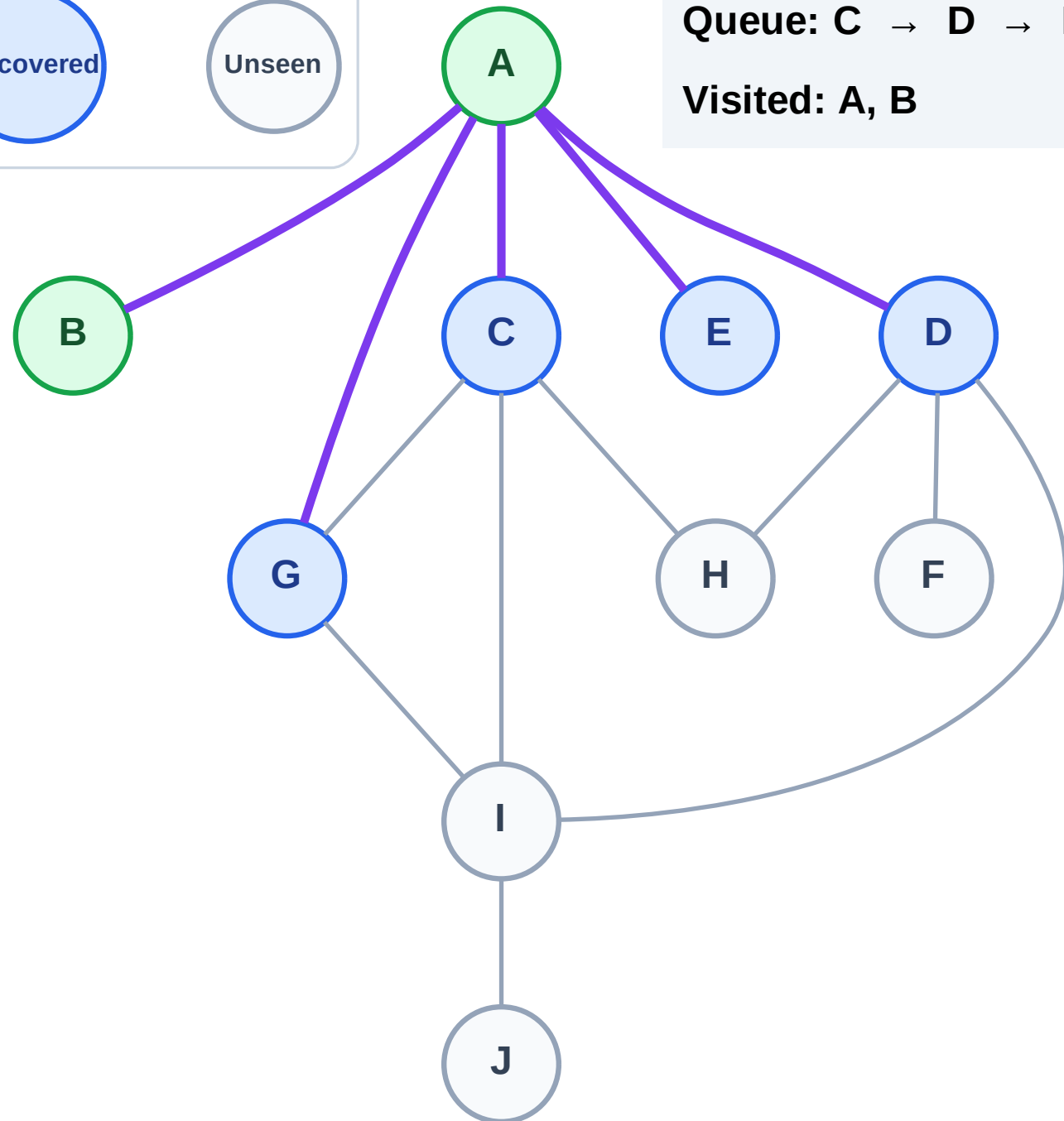
Step 5: No new neighbors from B

Legend



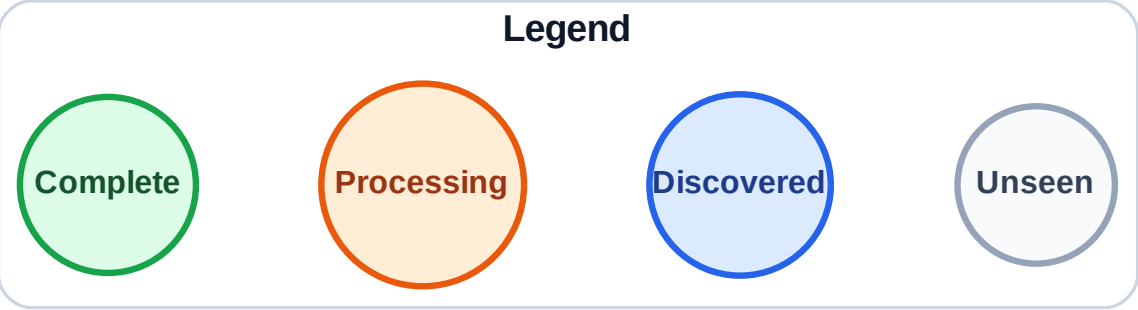
Queue: C → D → E → G

Visited: A, B

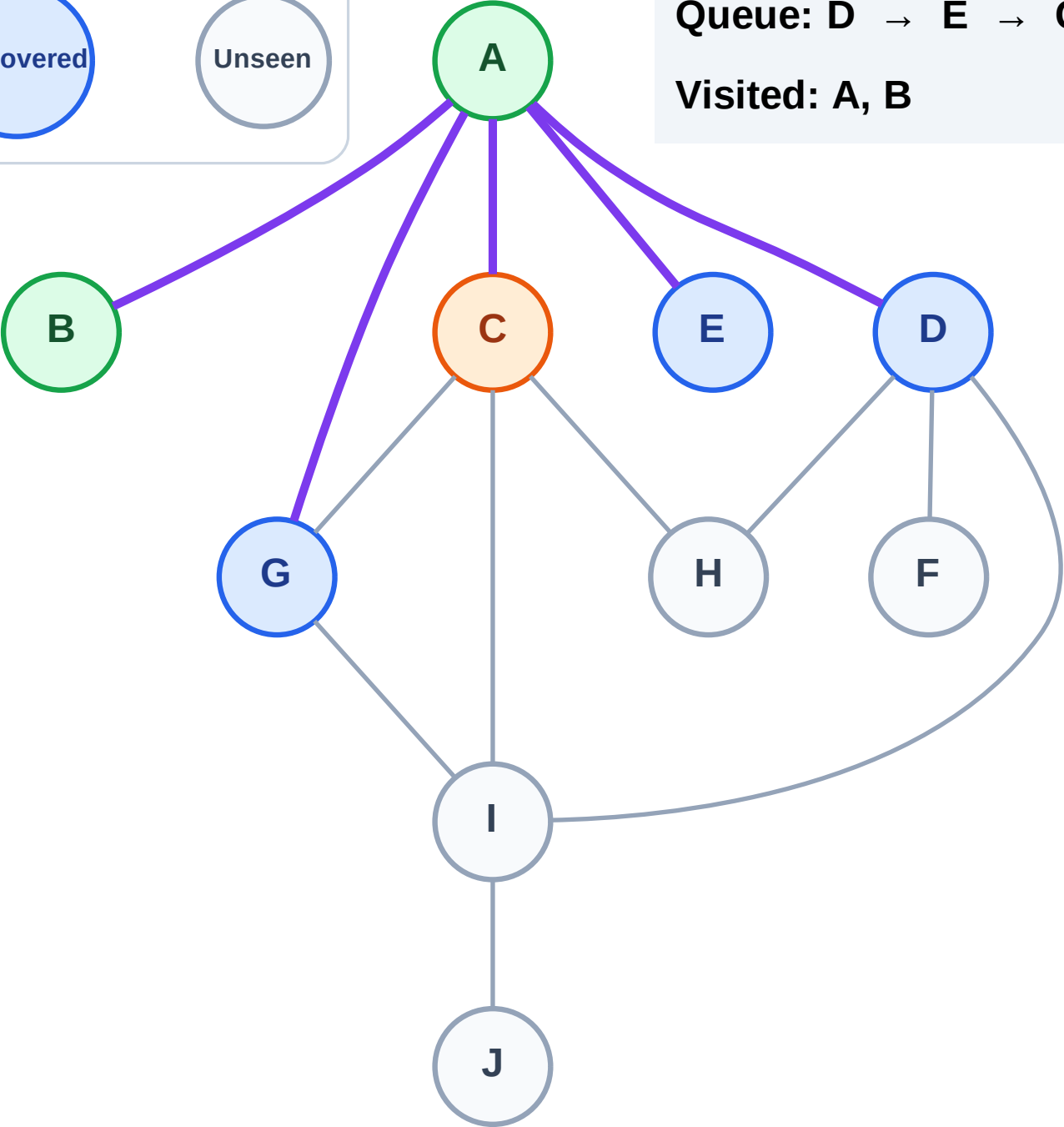


BFS Traversal

Step 6: Process node C



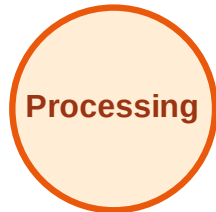
Queue: D → E → G
Visited: A, B



BFS Traversal

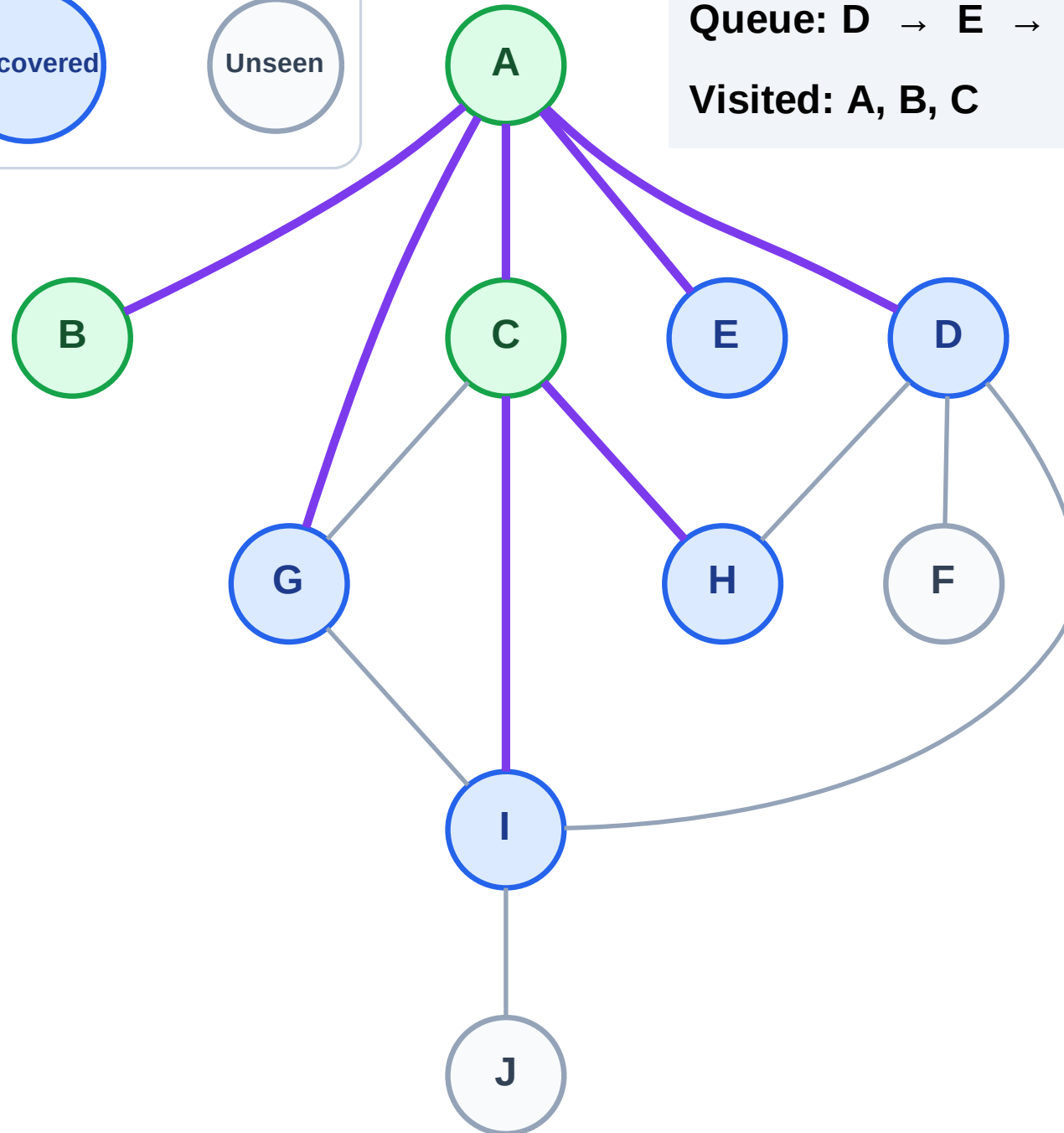
Step 7: Discover H, I from C

Legend



Queue: D → E → G → H → I

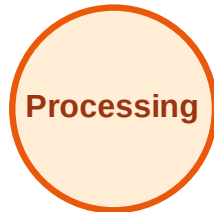
Visited: A, B, C



BFS Traversal

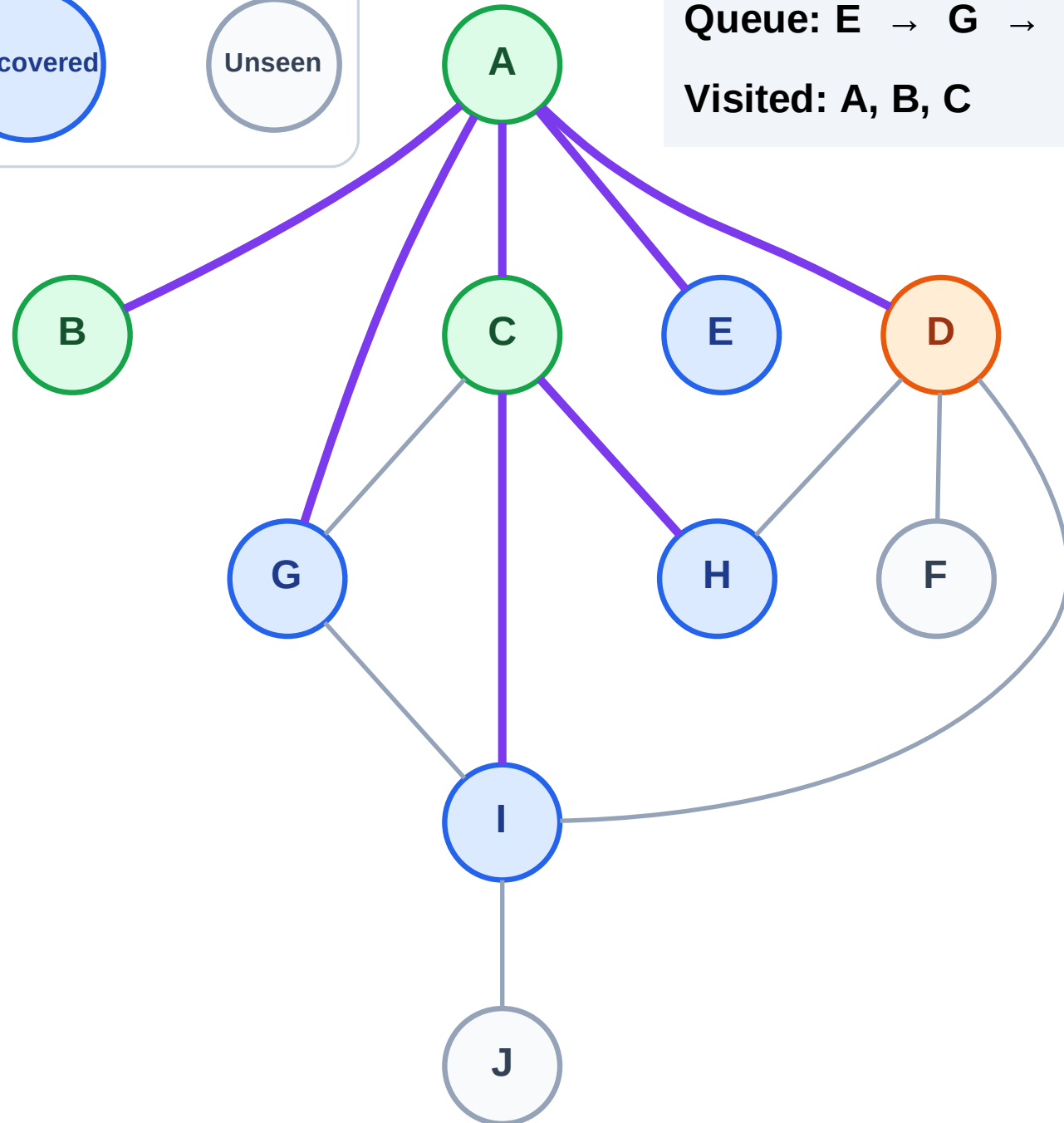
Step 8: Process node D

Legend



Queue: E → G → H → I

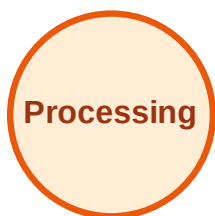
Visited: A, B, C



BFS Traversal

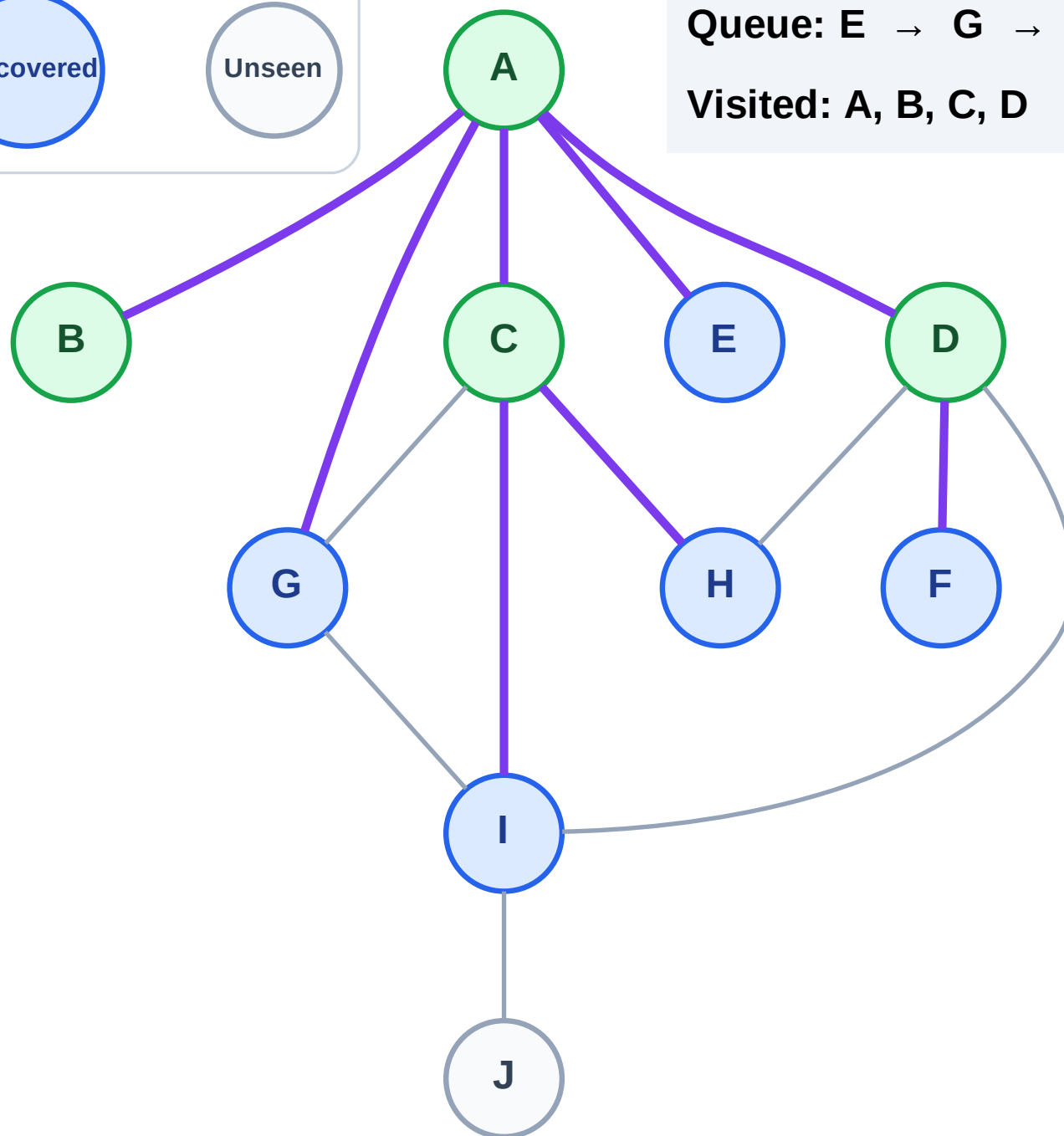
Step 9: Discover F from D

Legend



Queue: E → G → H → I → F

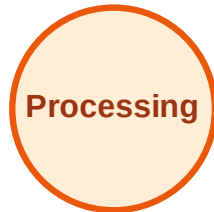
Visited: A, B, C, D



BFS Traversal

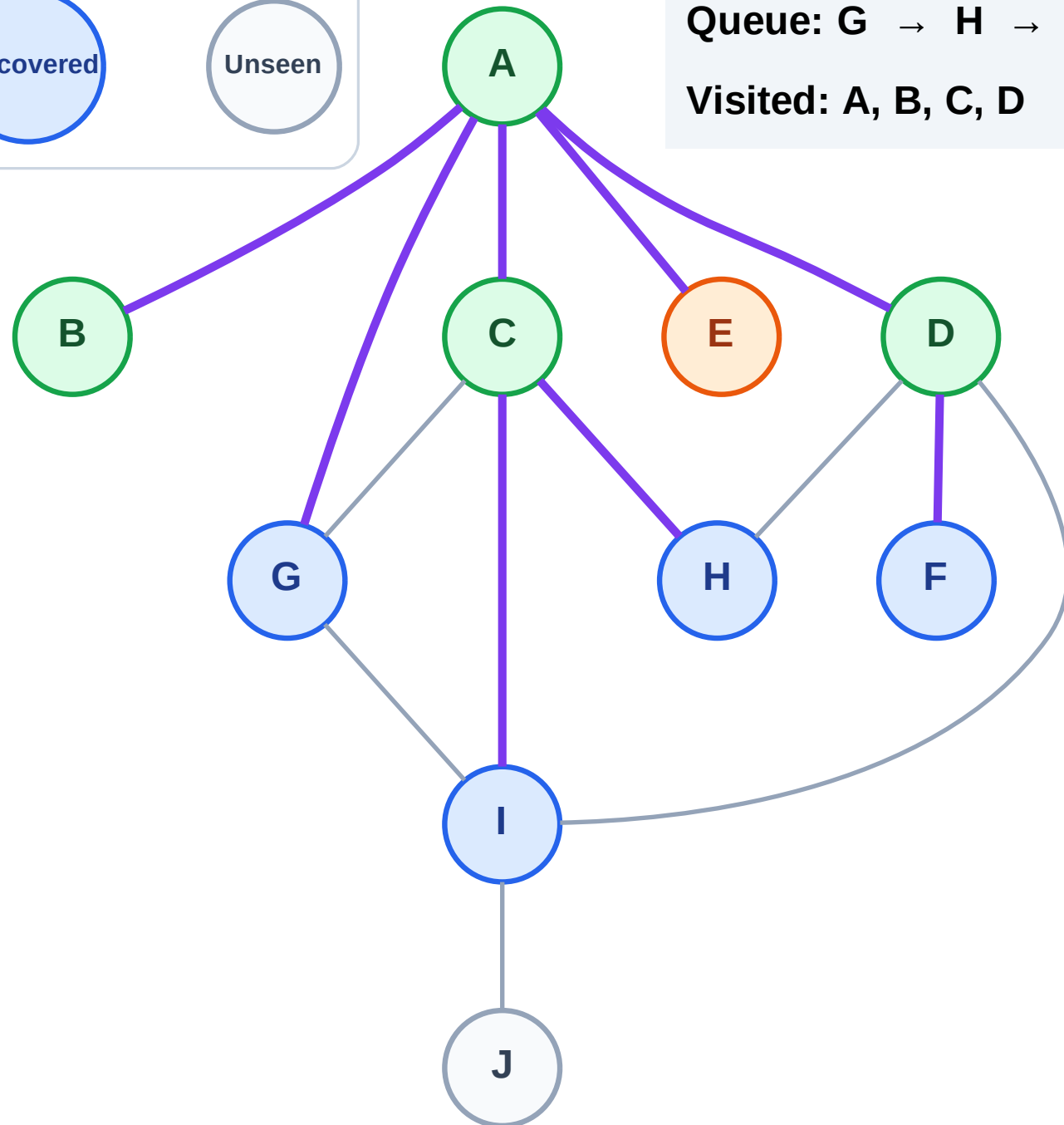
Step 10: Process node E

Legend



Queue: G → H → I → F

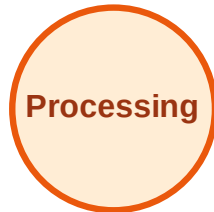
Visited: A, B, C, D



BFS Traversal

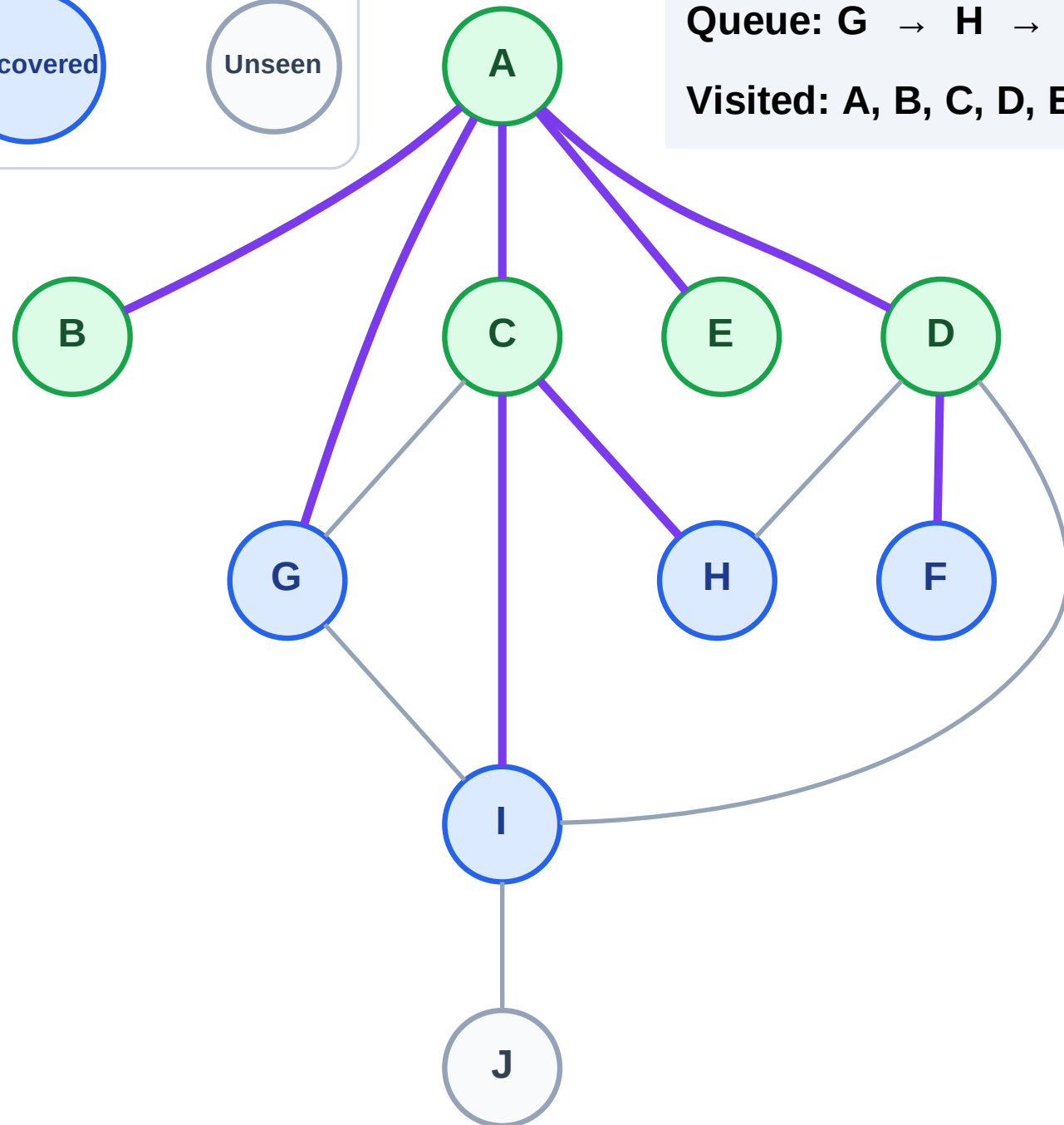
Step 11: No new neighbors from E

Legend



Queue: G → H → I → F

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node G

Legend

Complete

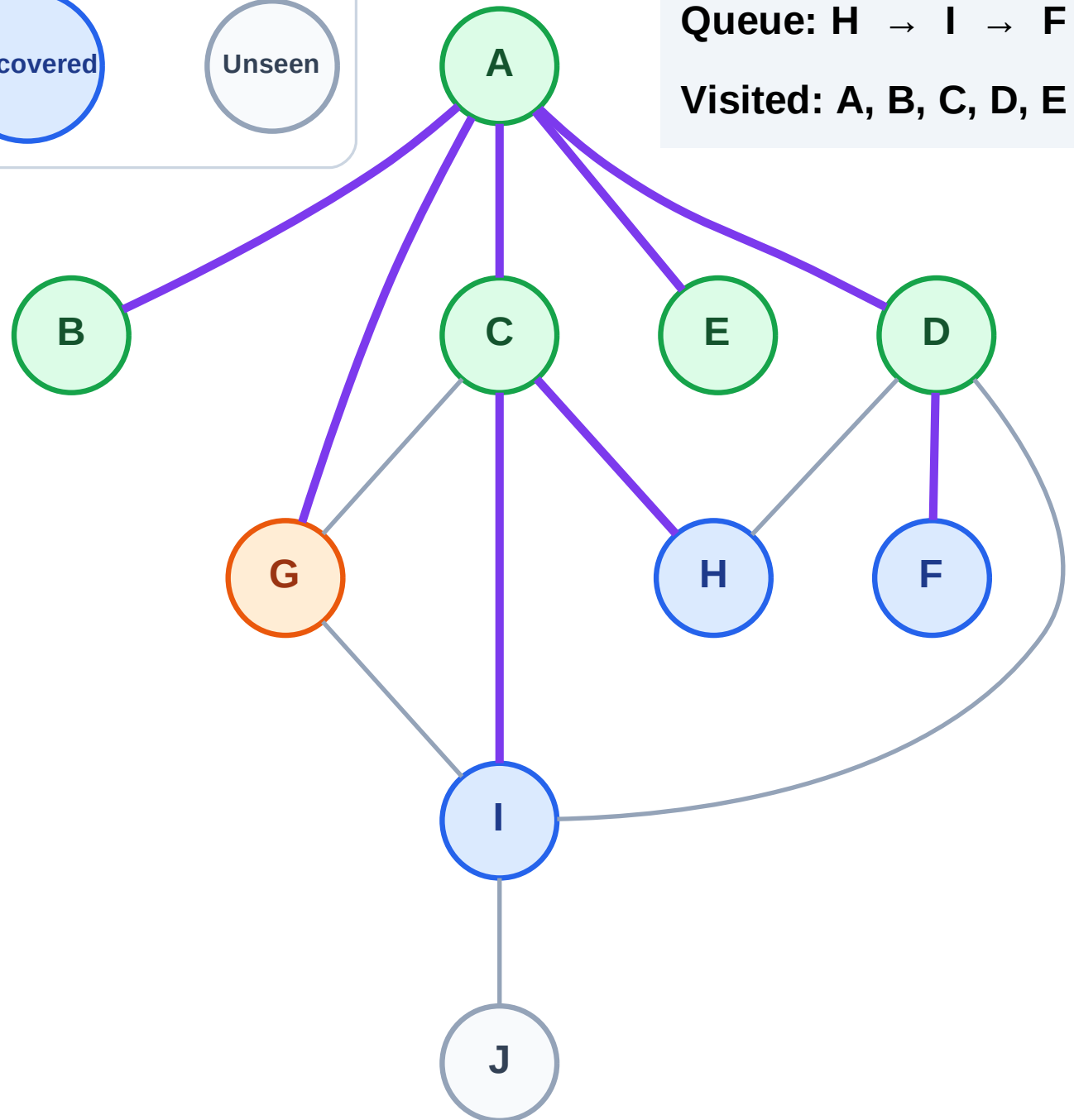
Processing

Discovered

Unseen

Queue: H → I → F

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

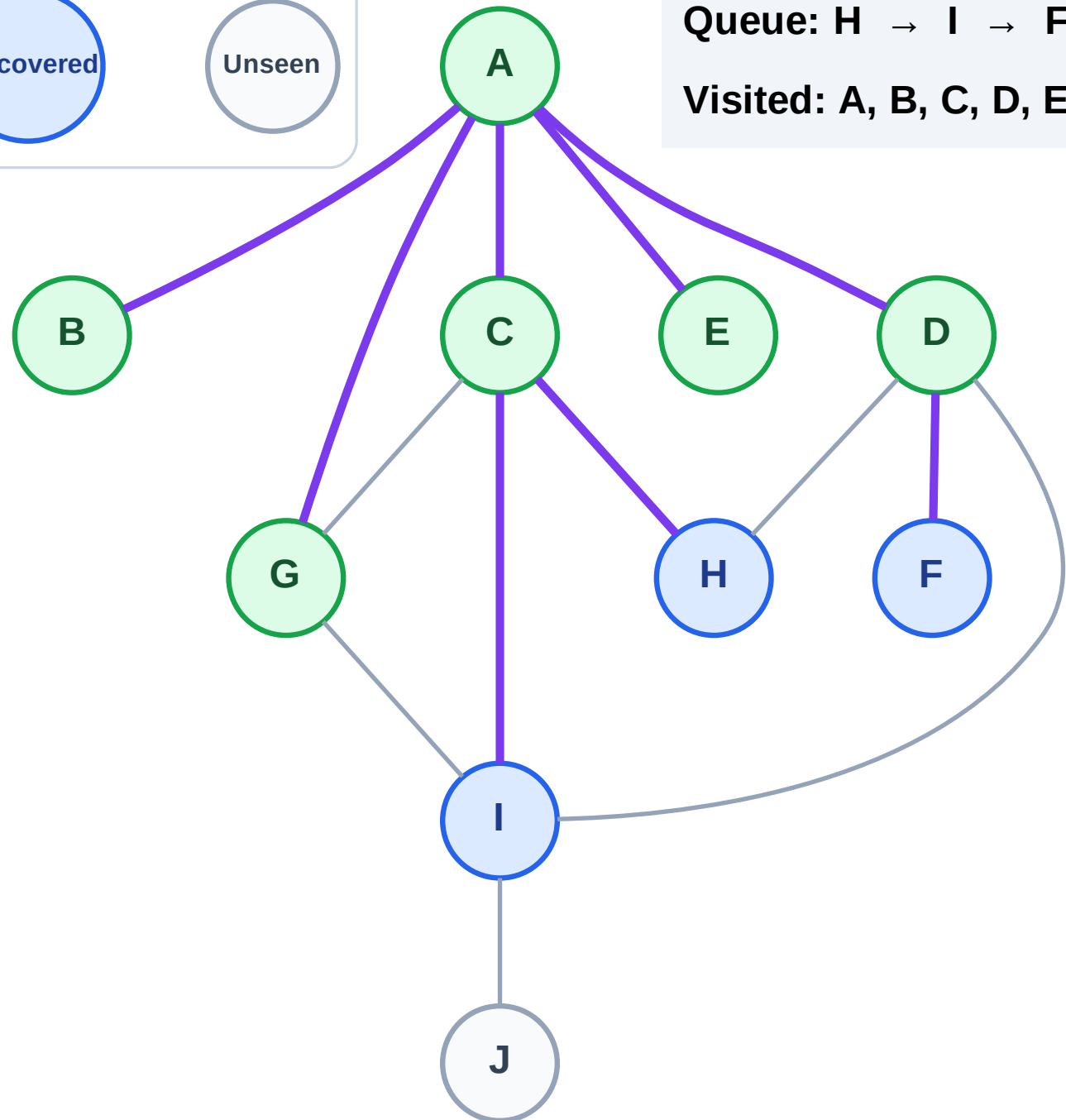
Processing

Discovered

Unseen

Queue: H → I → F

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node H

Legend

Complete

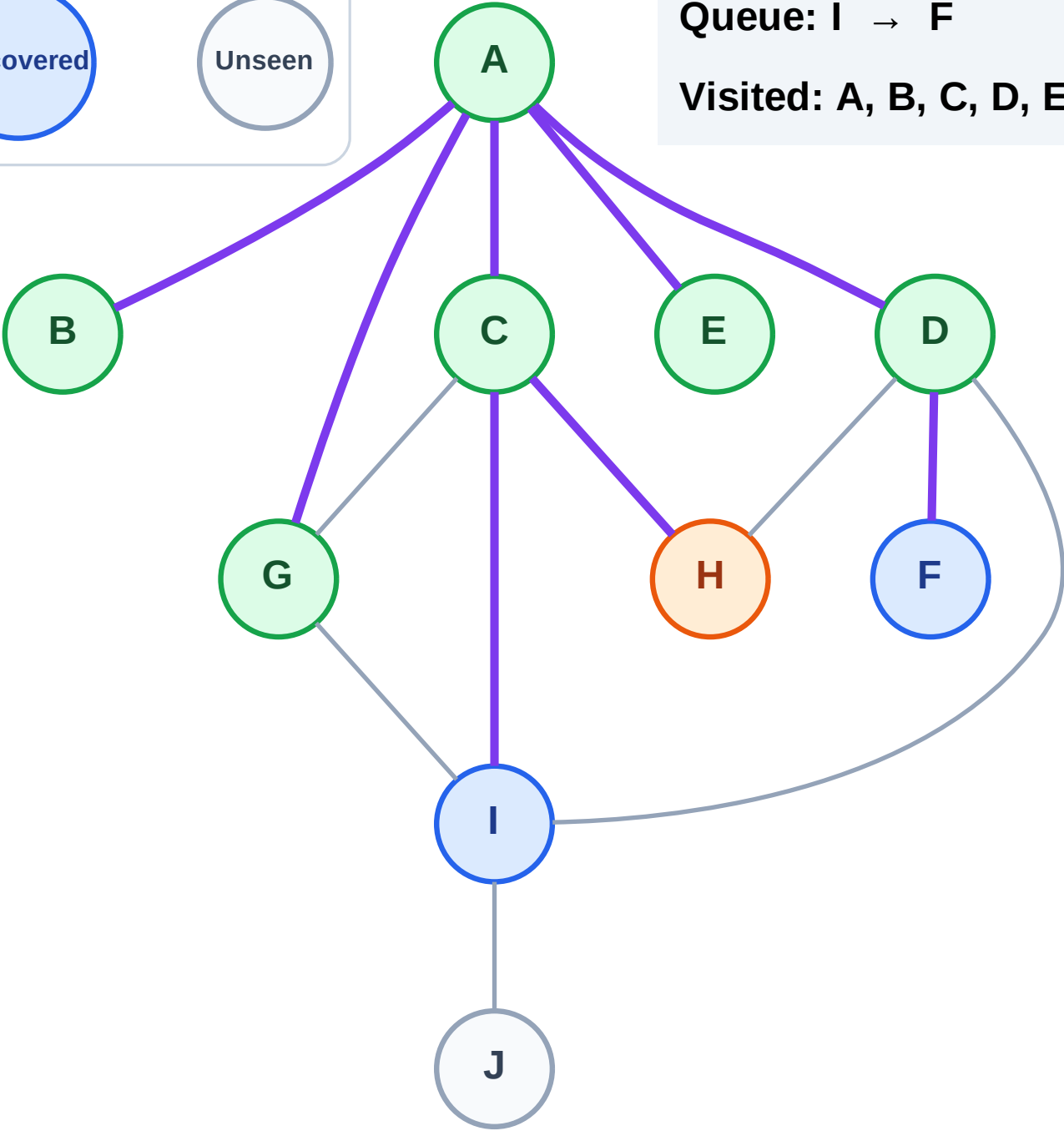
Processing

Discovered

Unseen

Queue: I → F

Visited: A, B, C, D, E, G



BFS Traversal

Step 15: No new neighbors from H

Legend

Complete

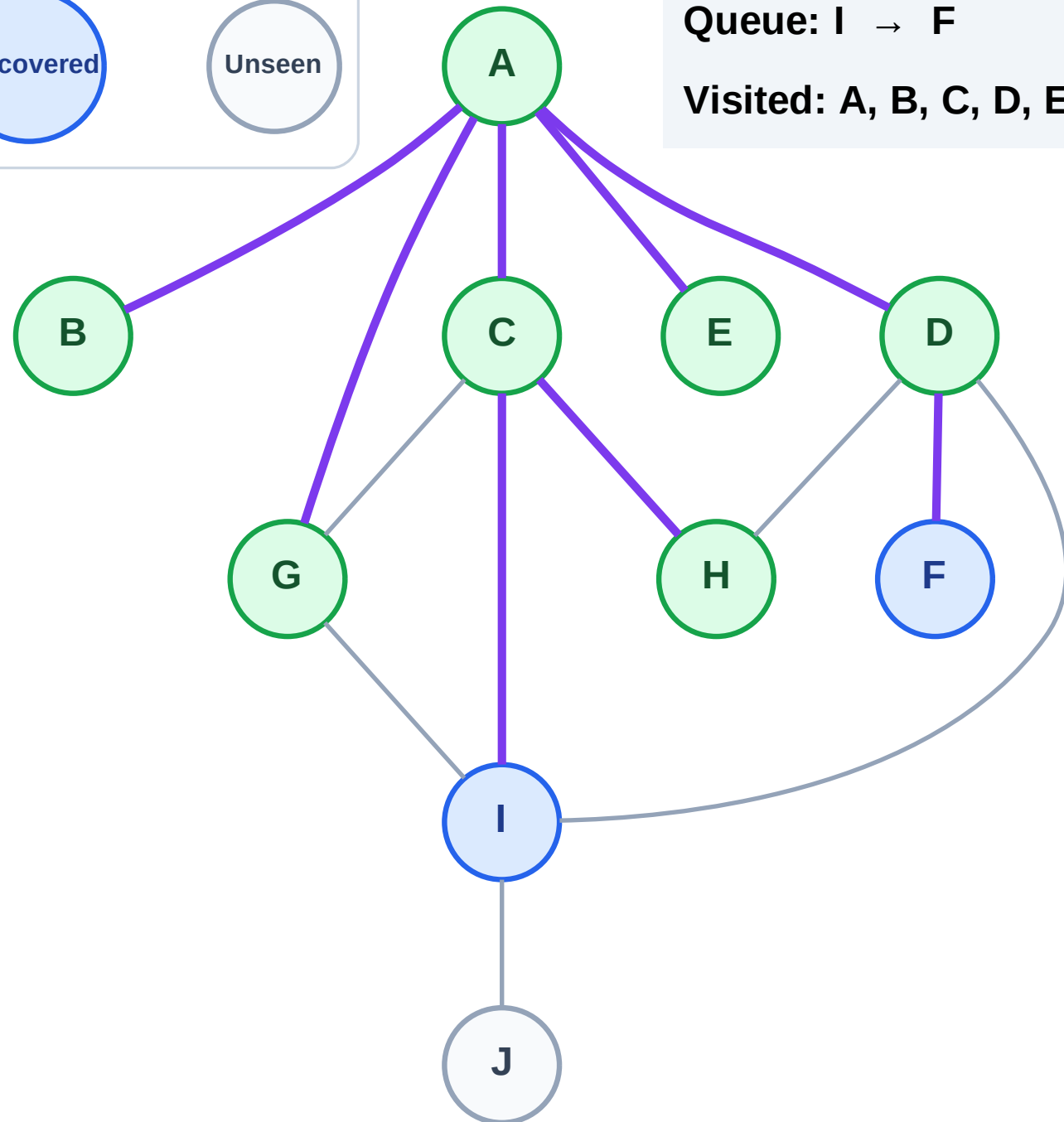
Processing

Discovered

Unseen

Queue: I → F

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node I

Legend

Complete

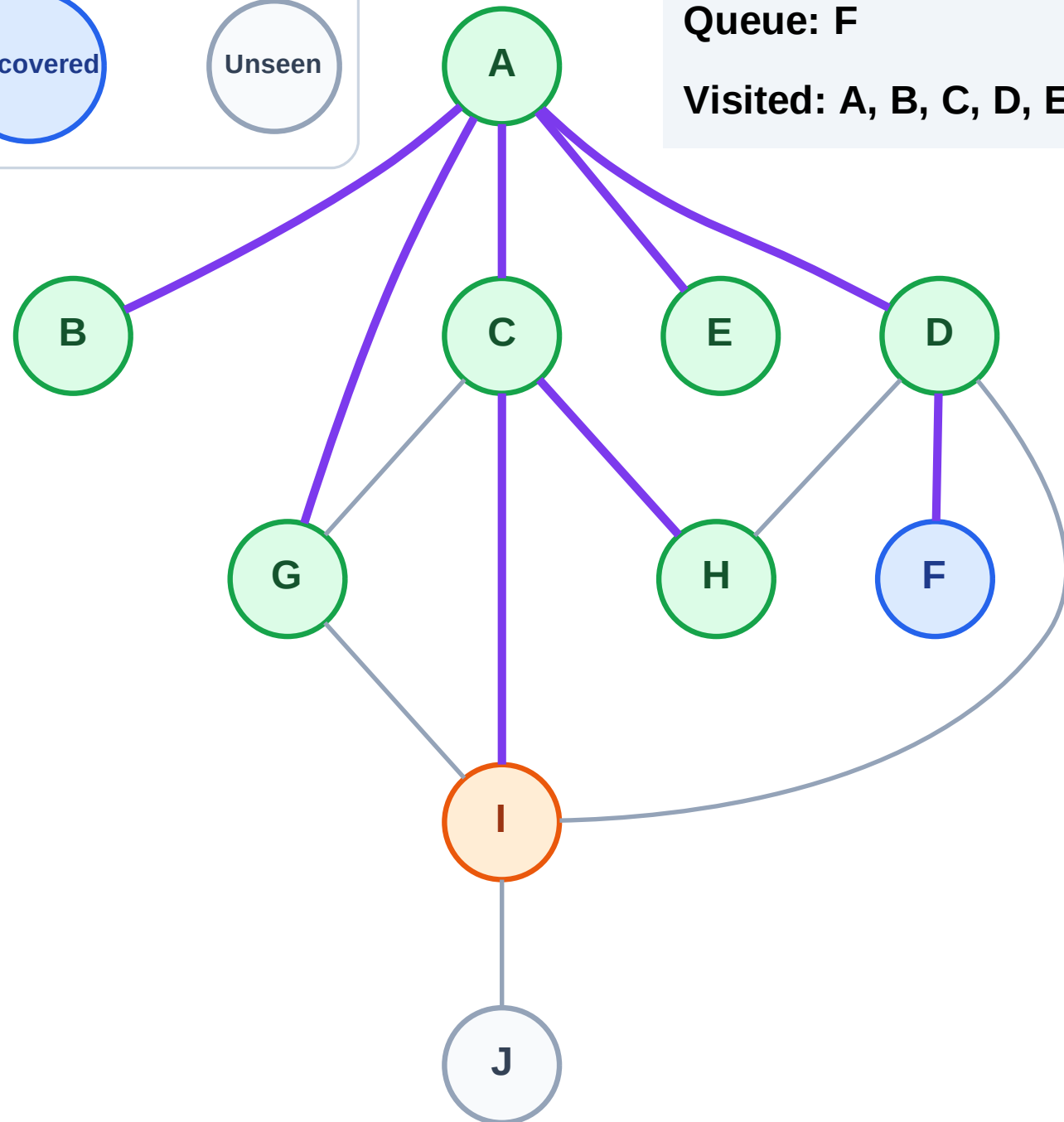
Processing

Discovered

Unseen

Queue: F

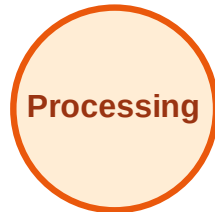
Visited: A, B, C, D, E, G, H



BFS Traversal

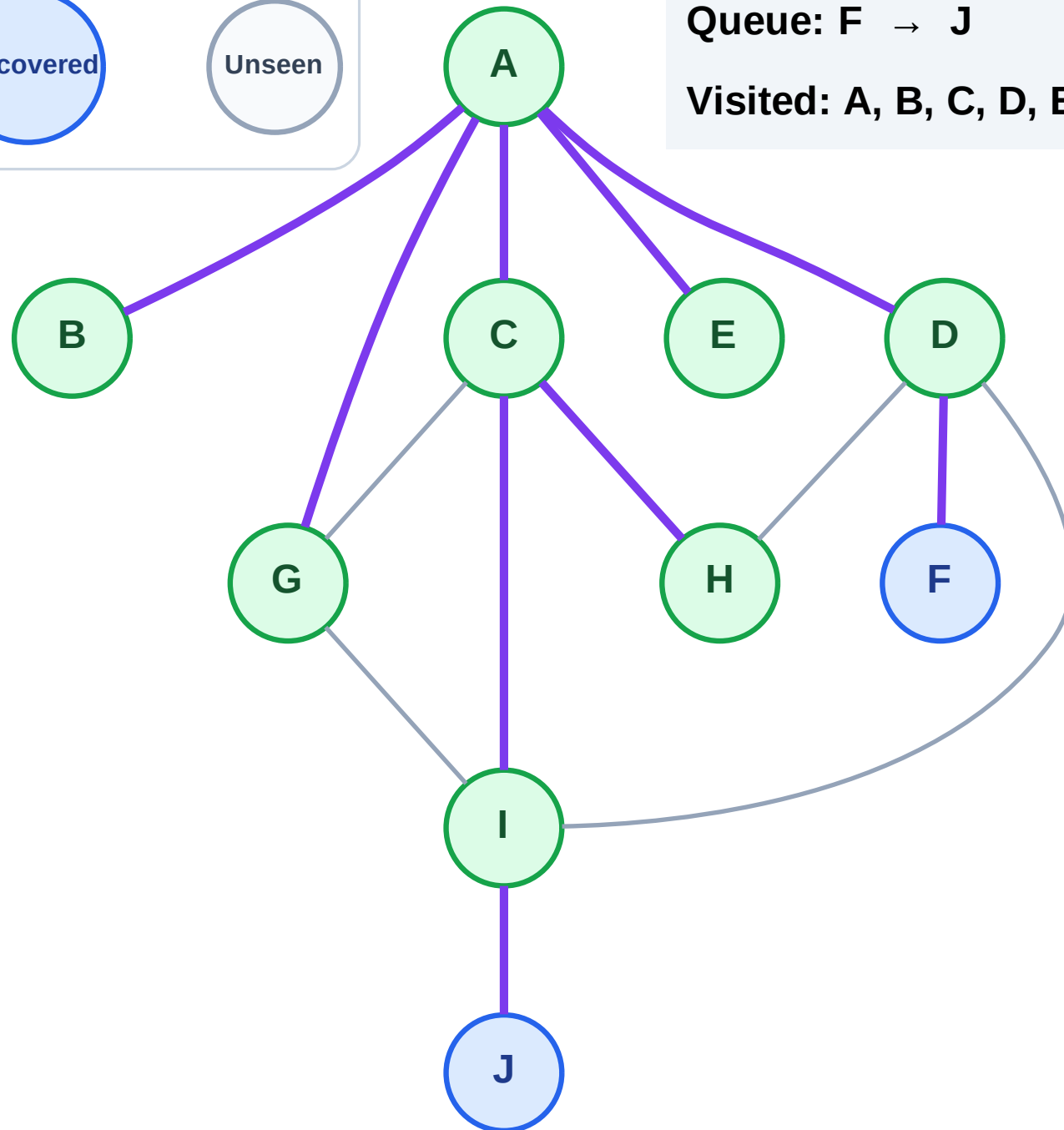
Step 17: Discover J from I

Legend



Queue: F → J

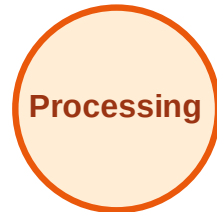
Visited: A, B, C, D, E, G, H, I



BFS Traversal

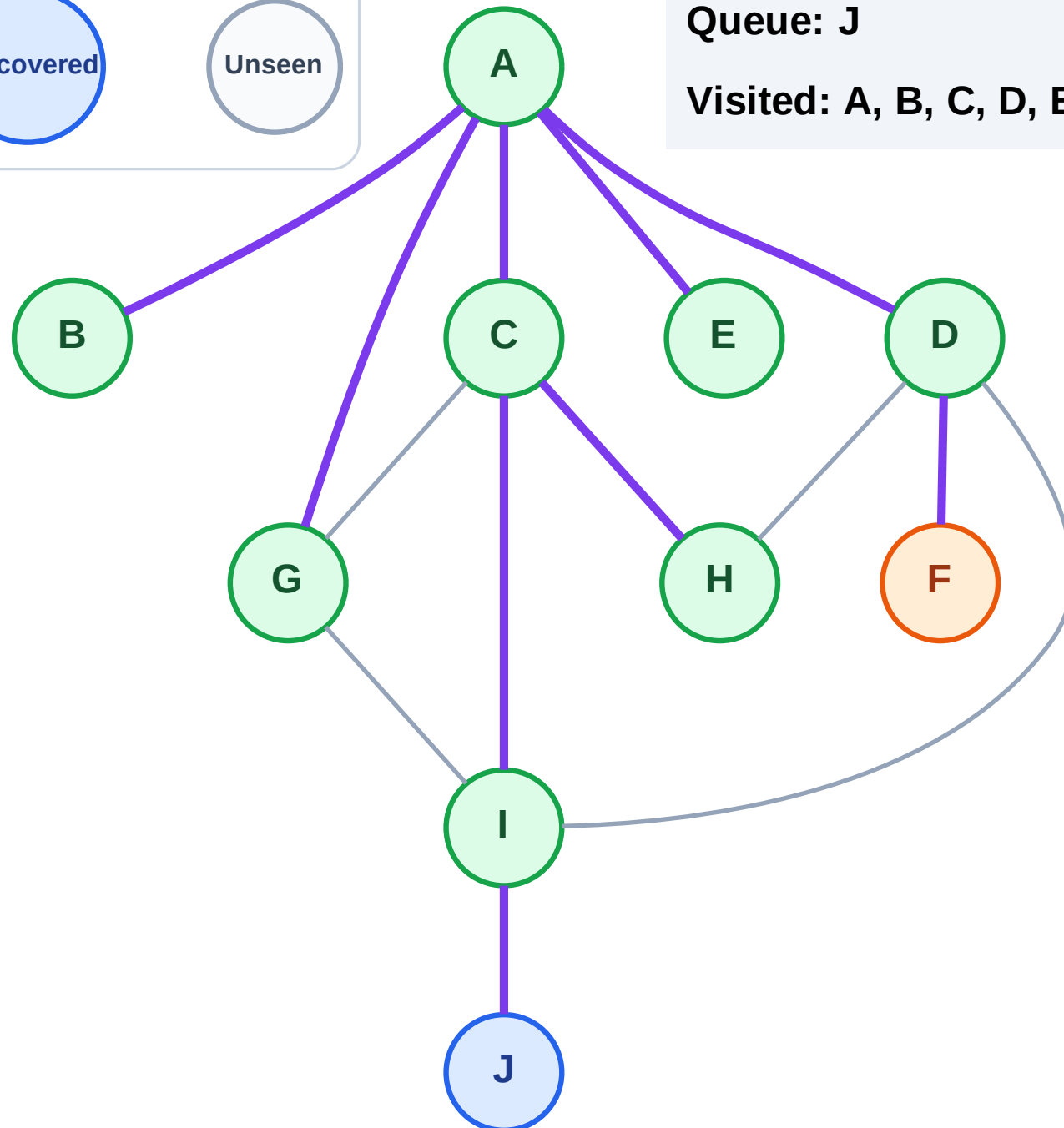
Step 18: Process node F

Legend



Queue: J

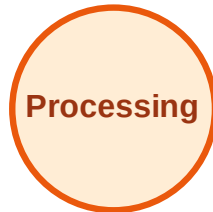
Visited: A, B, C, D, E, G, H, I



BFS Traversal

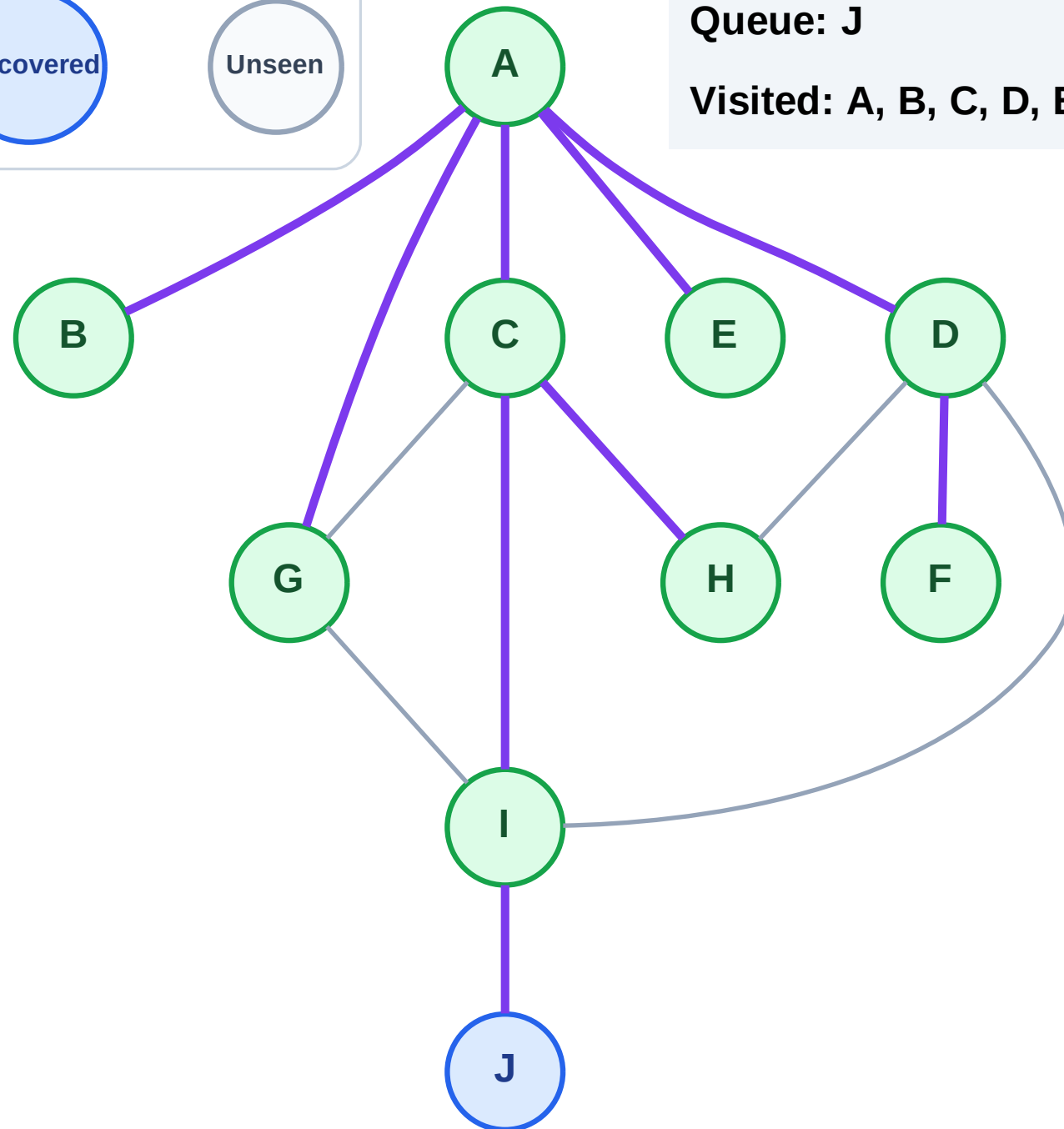
Step 19: No new neighbors from F

Legend



Queue: J

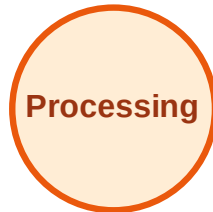
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

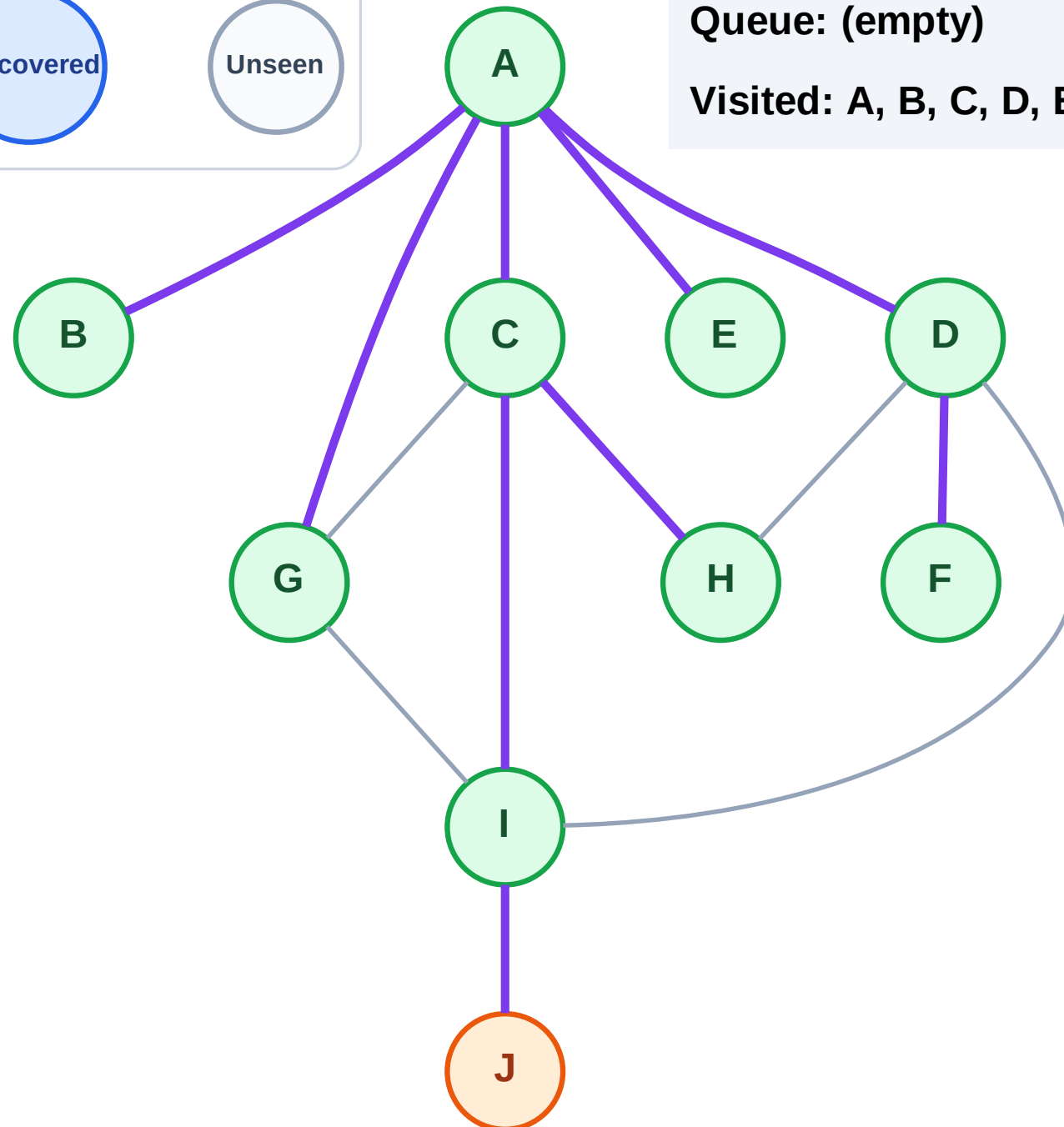
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

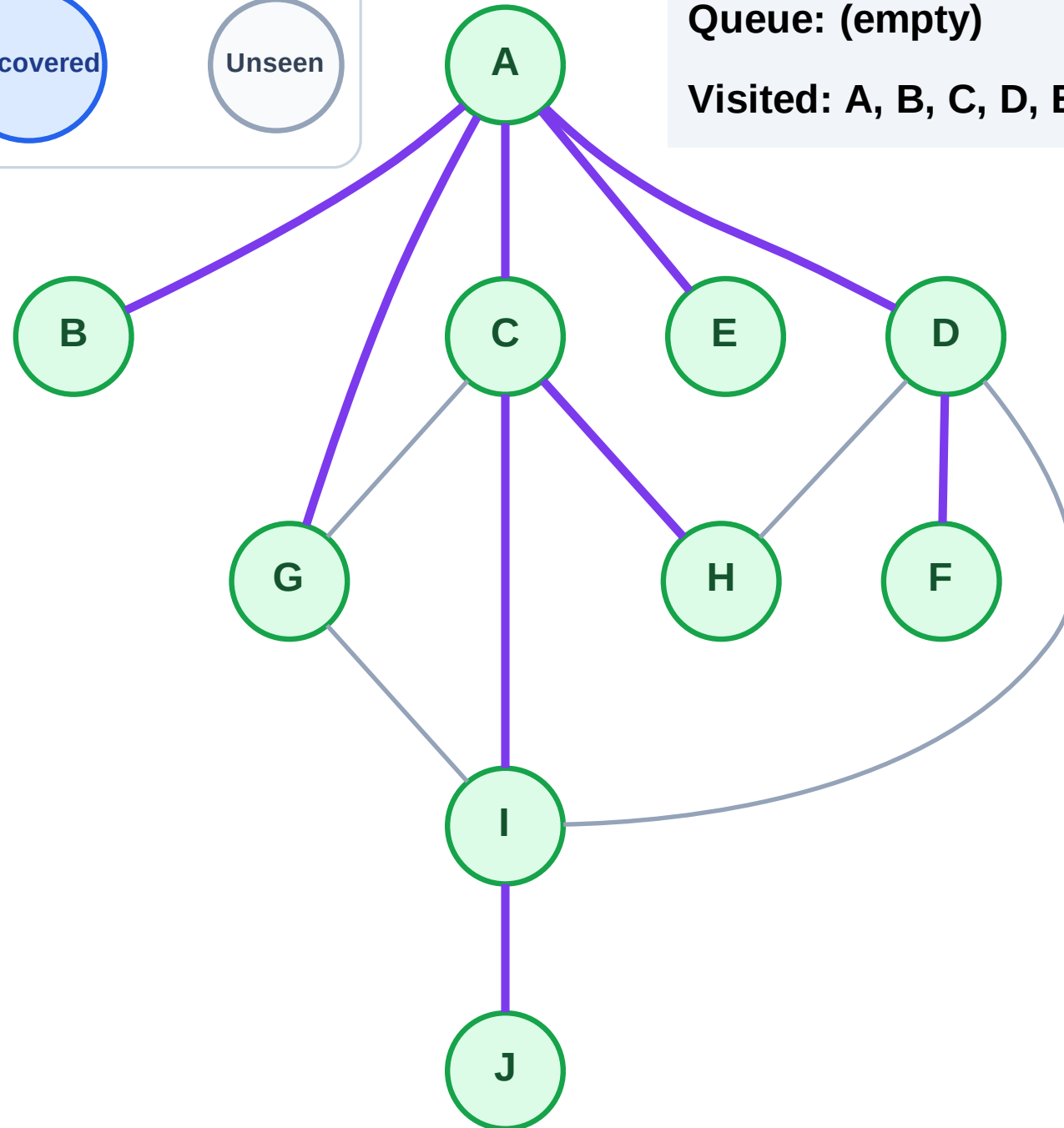
Step 21: No new neighbors from J

Legend



Queue: (empty)

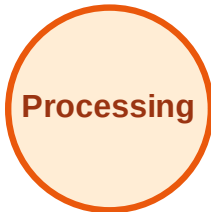
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

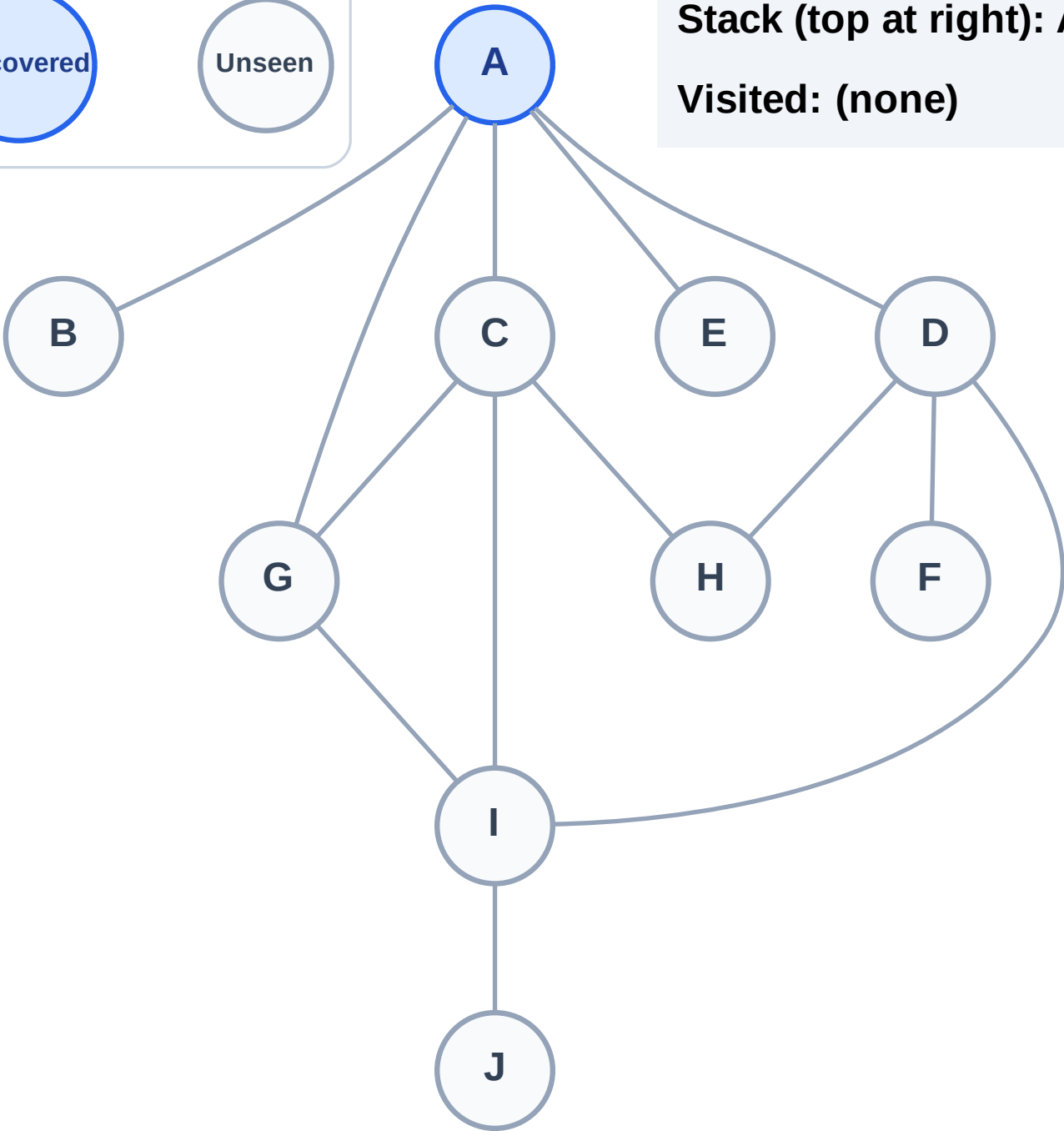
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

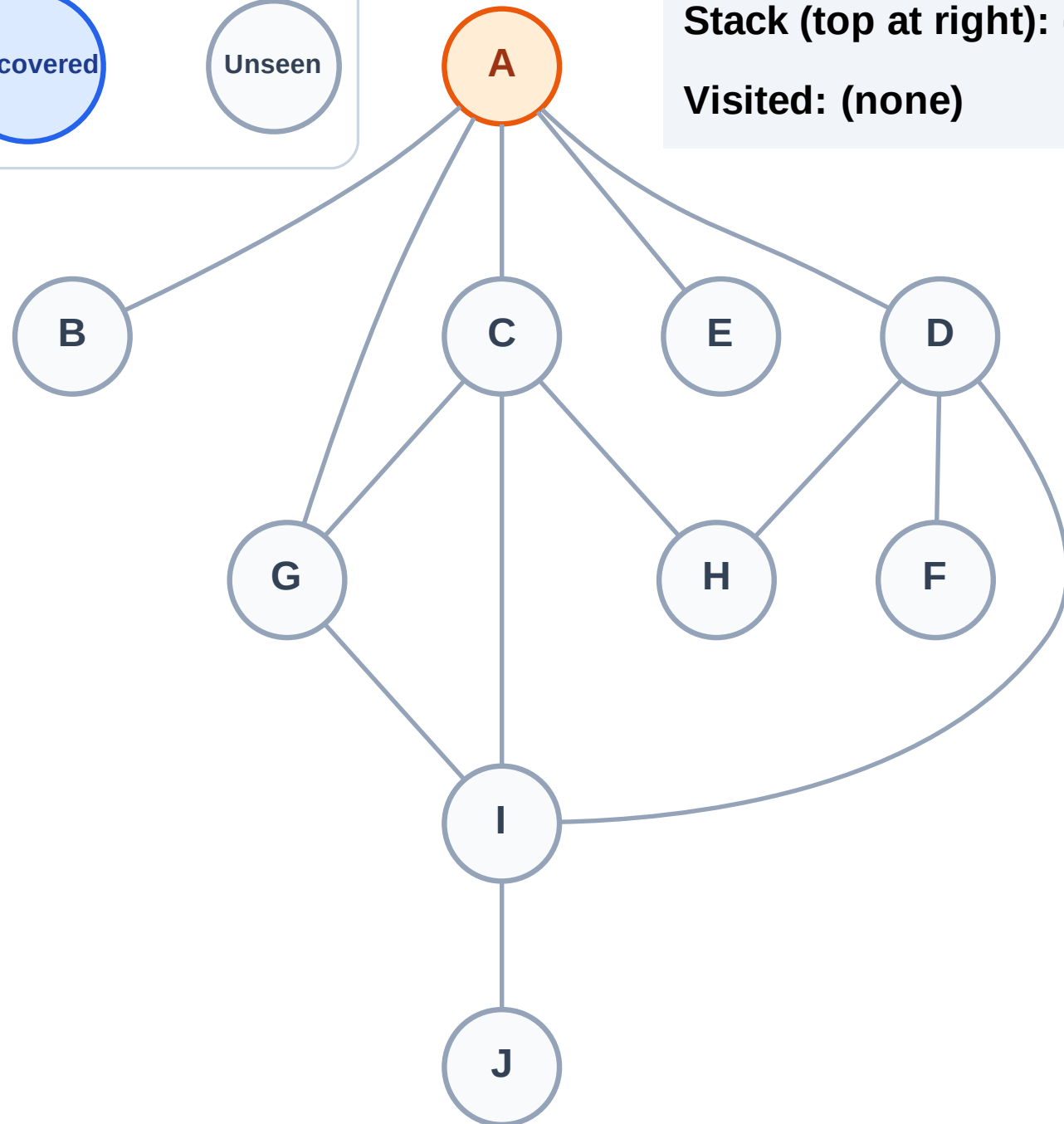
Processing

Discovered

Unseen

Stack (top at right): (empty)

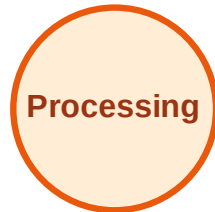
Visited: (none)



DFS Traversal

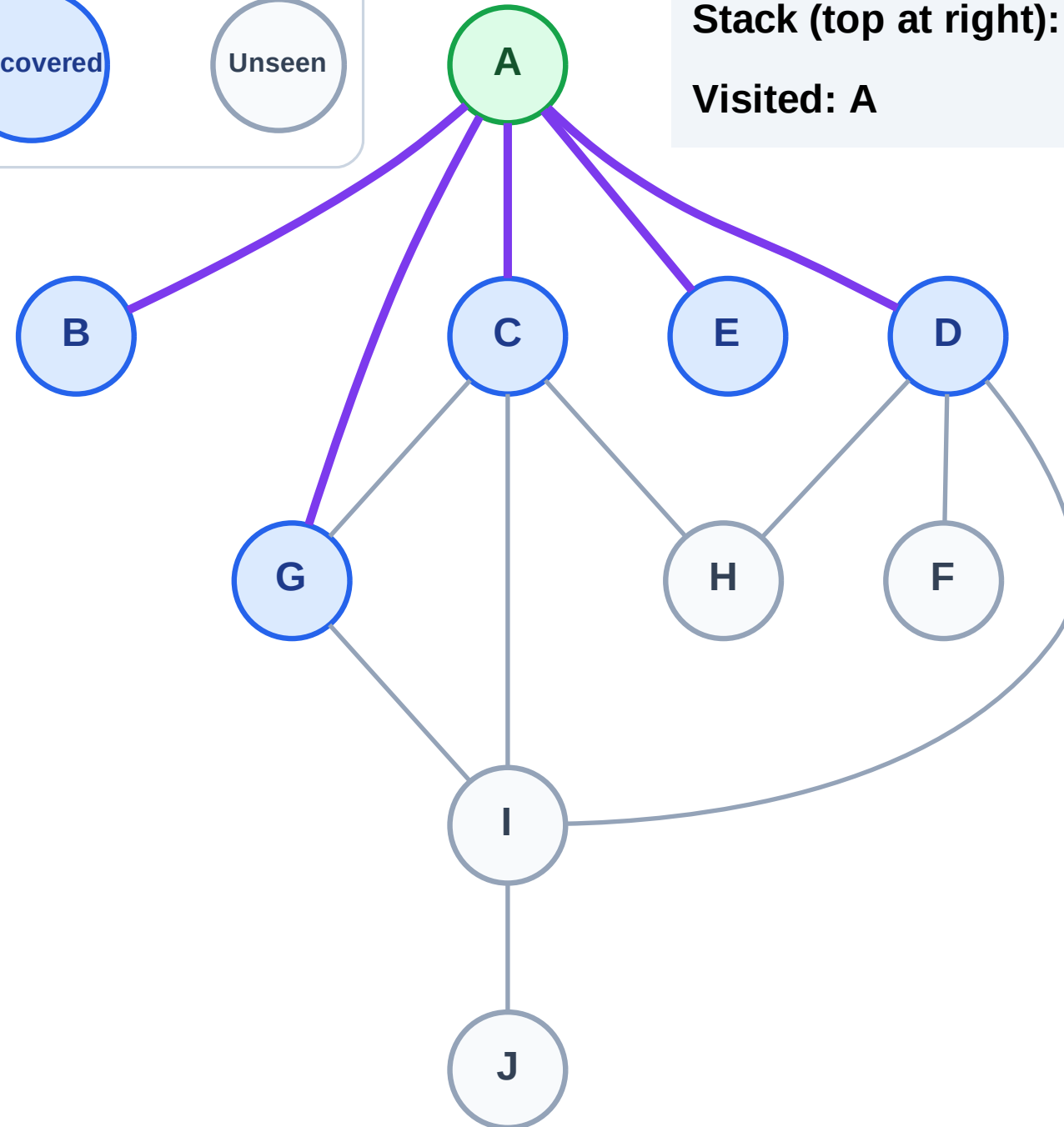
Step 3: Discover G, E, D, C, B from A

Legend



Stack (top at right): G | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

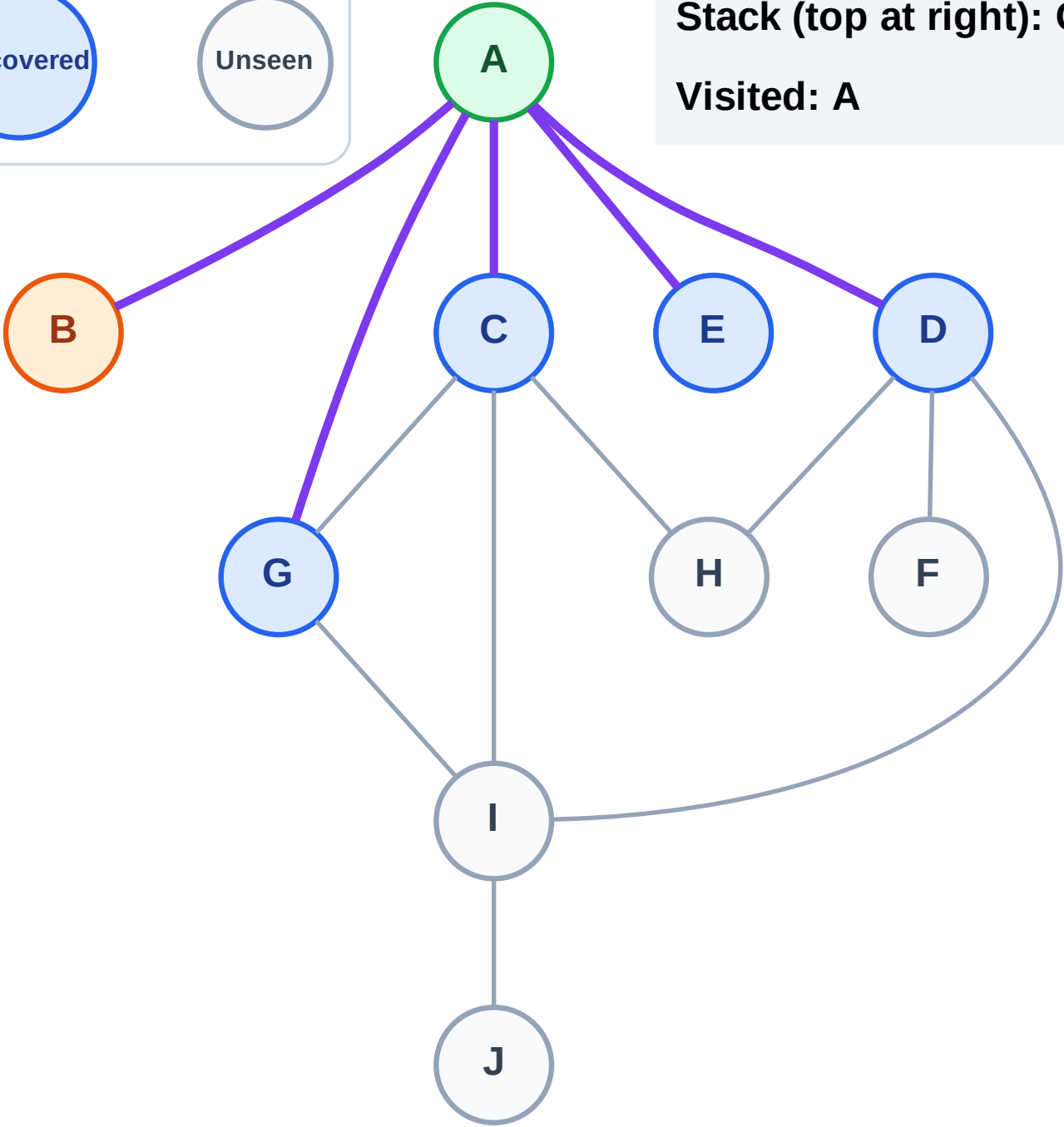
Processing

Discovered

Unseen

Stack (top at right): G | E | D | C

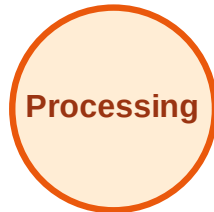
Visited: A



DFS Traversal

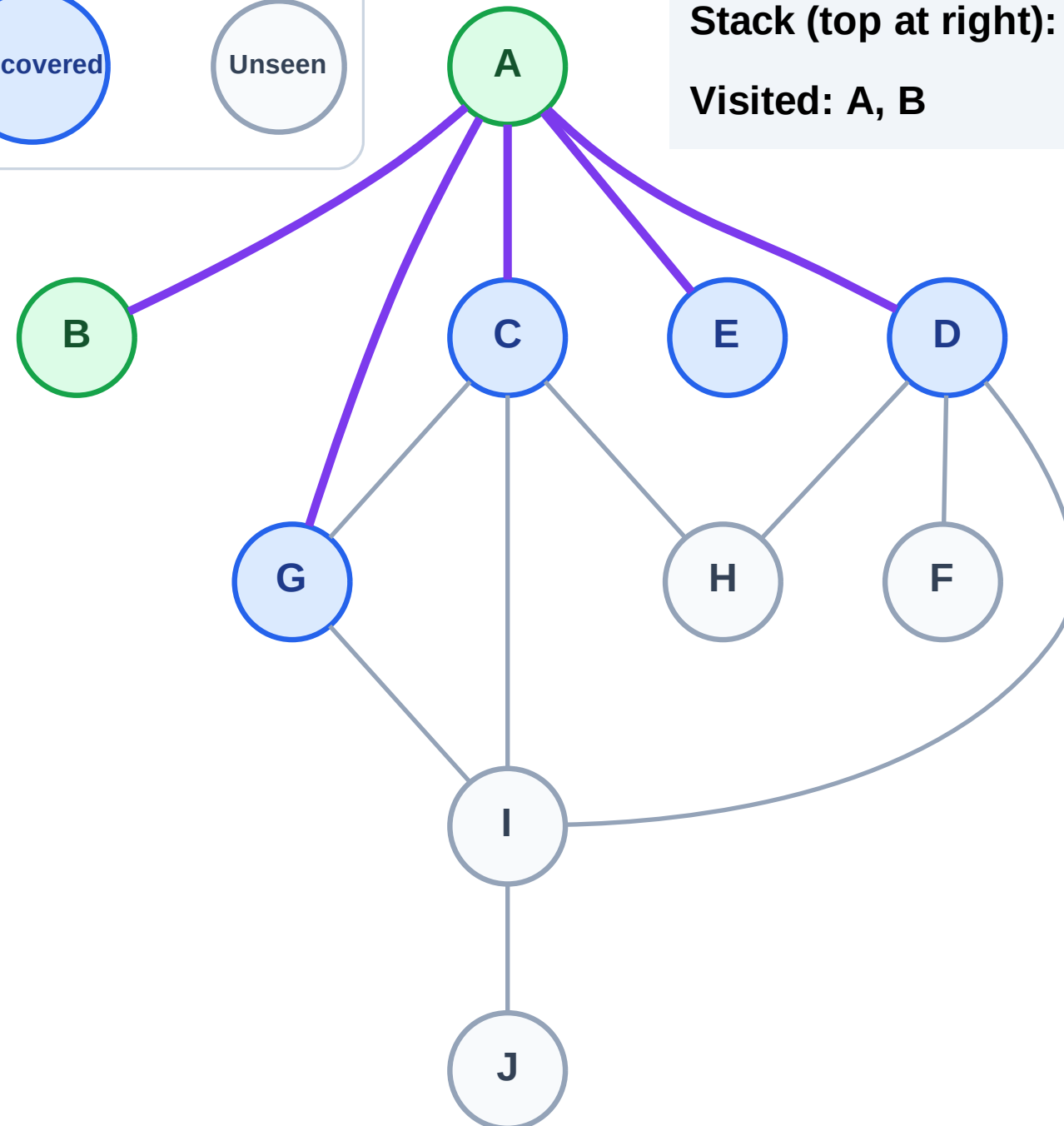
Step 5: No new neighbors from B

Legend



Stack (top at right): G | E | D | C

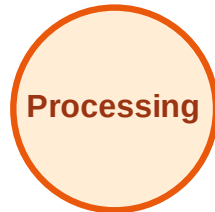
Visited: A, B



DFS Traversal

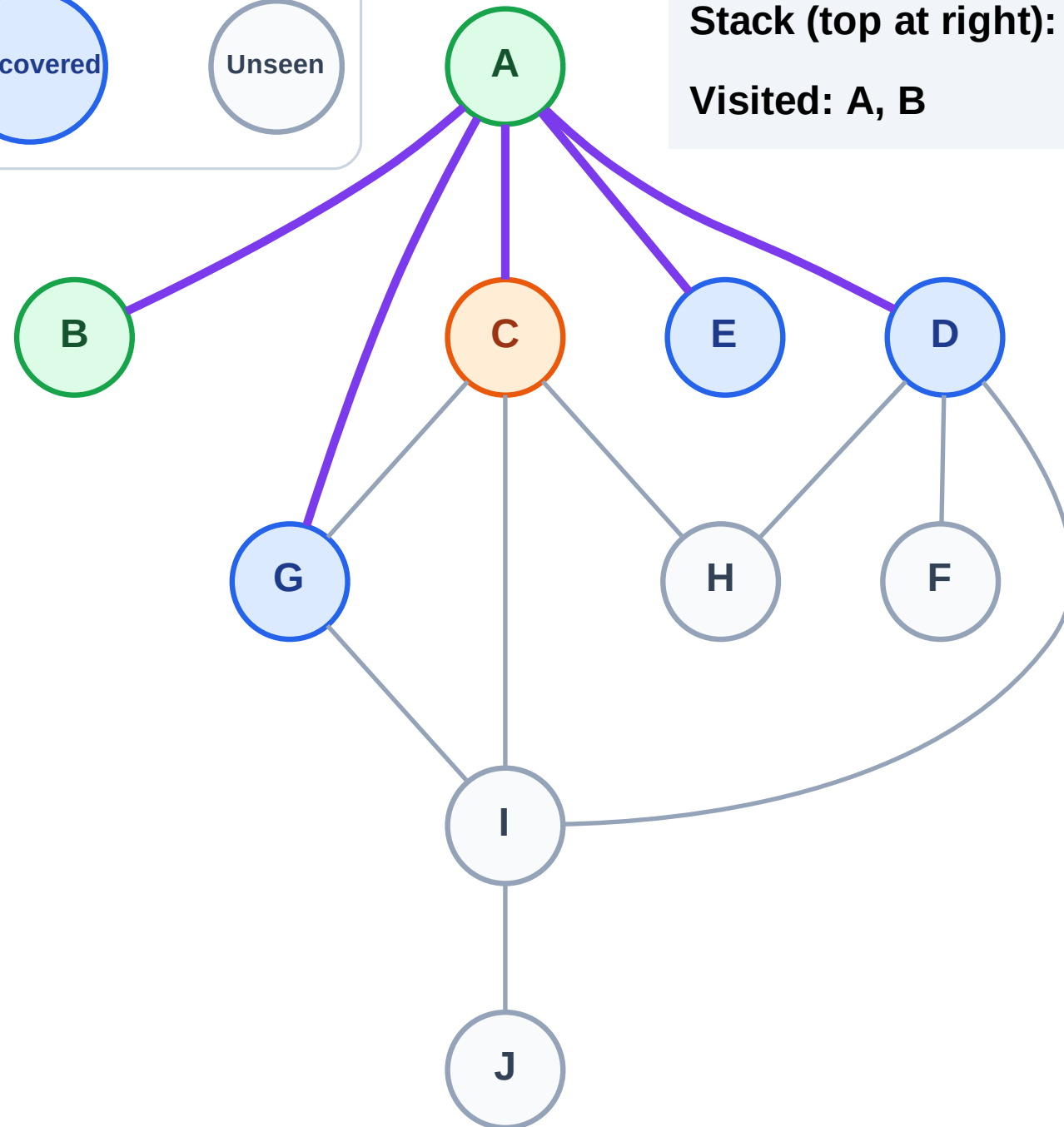
Step 6: Process node C

Legend



Stack (top at right): G | E | D

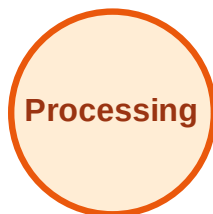
Visited: A, B



DFS Traversal

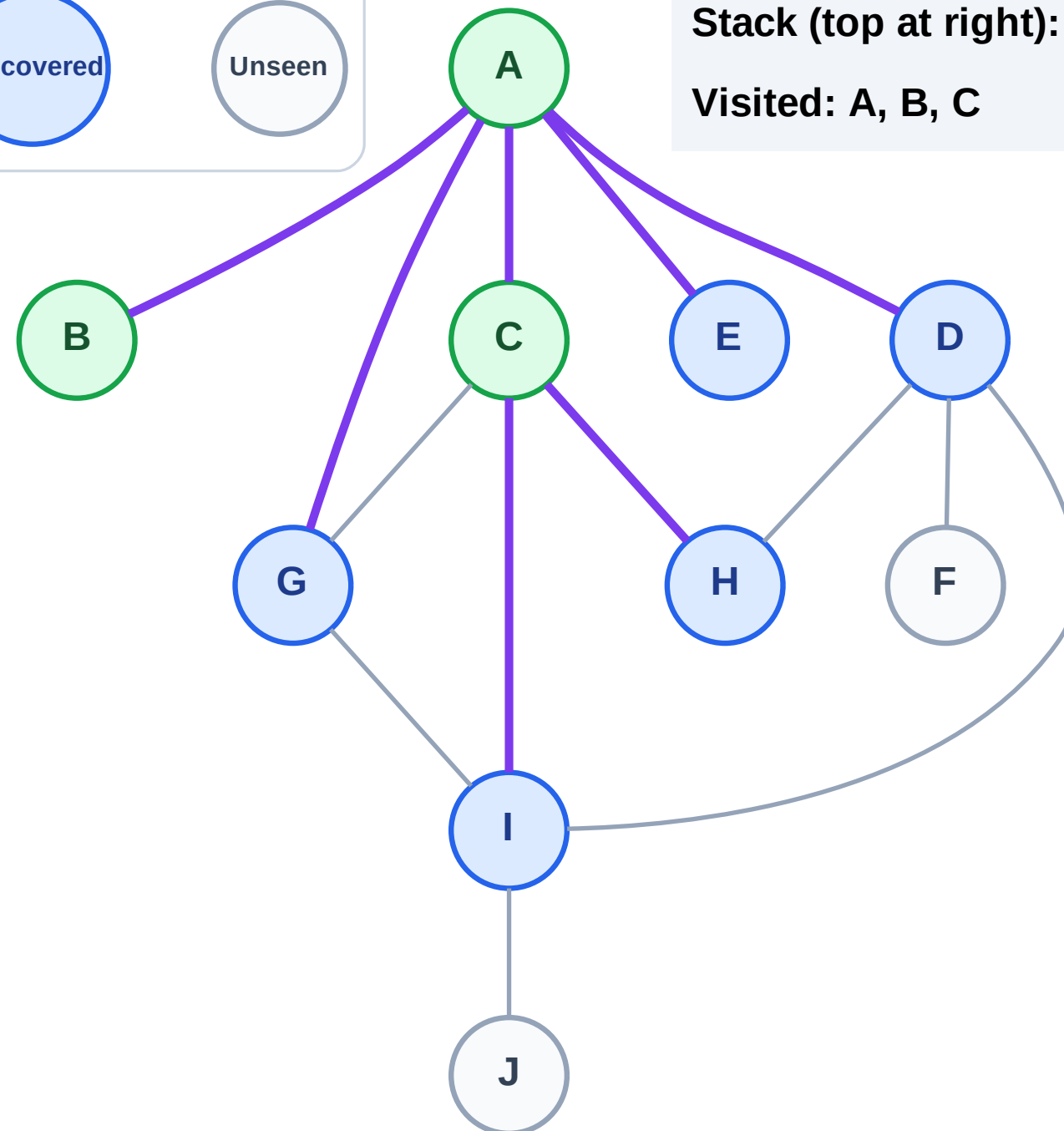
Step 7: Discover I, H from C

Legend



Stack (top at right): G | E | D | I | H

Visited: A, B, C



DFS Traversal

Step 8: Process node H

Legend

Complete

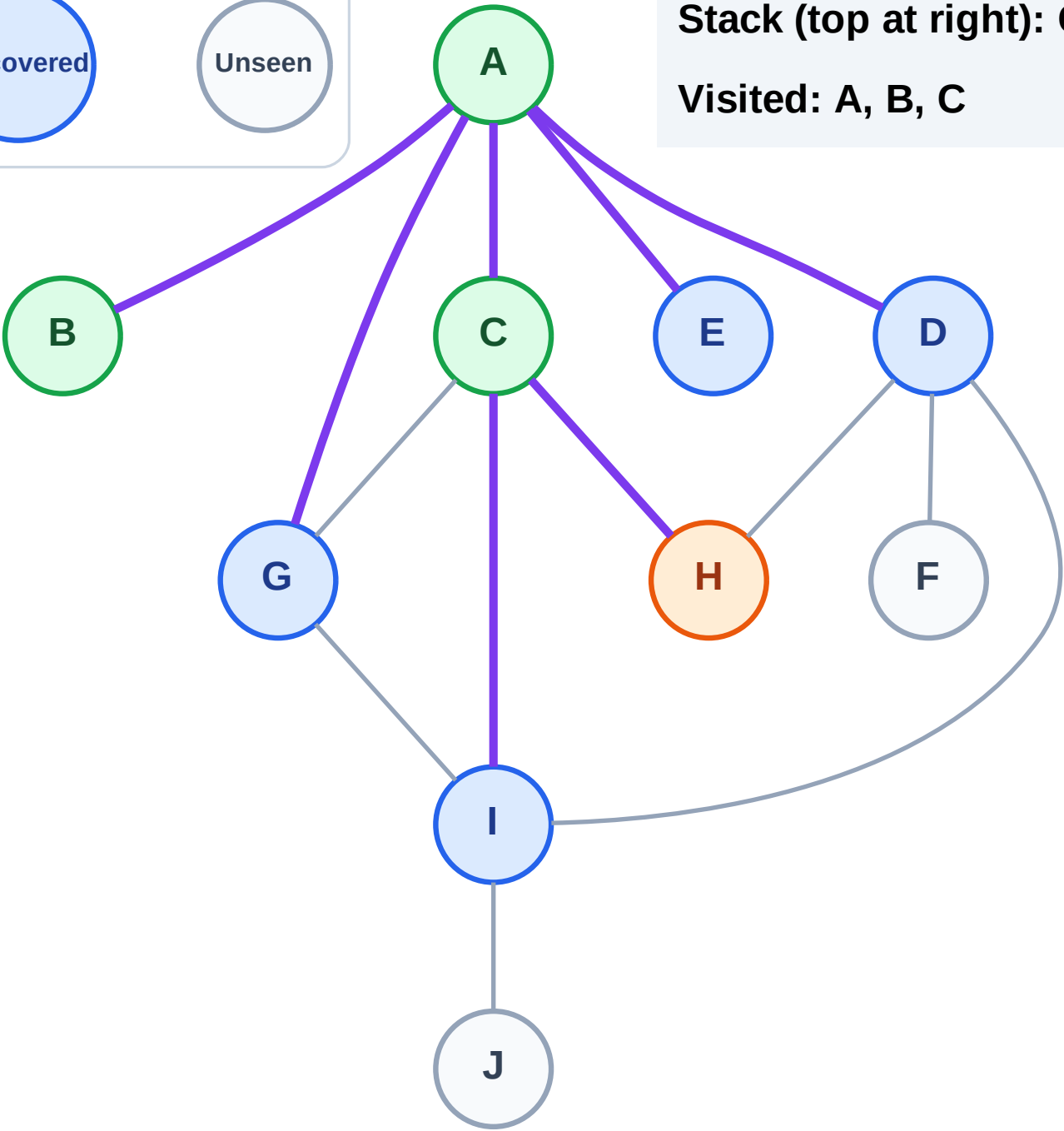
Processing

Discovered

Unseen

Stack (top at right): G | E | D | I

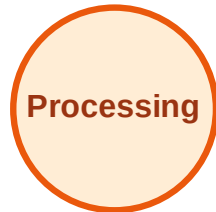
Visited: A, B, C



DFS Traversal

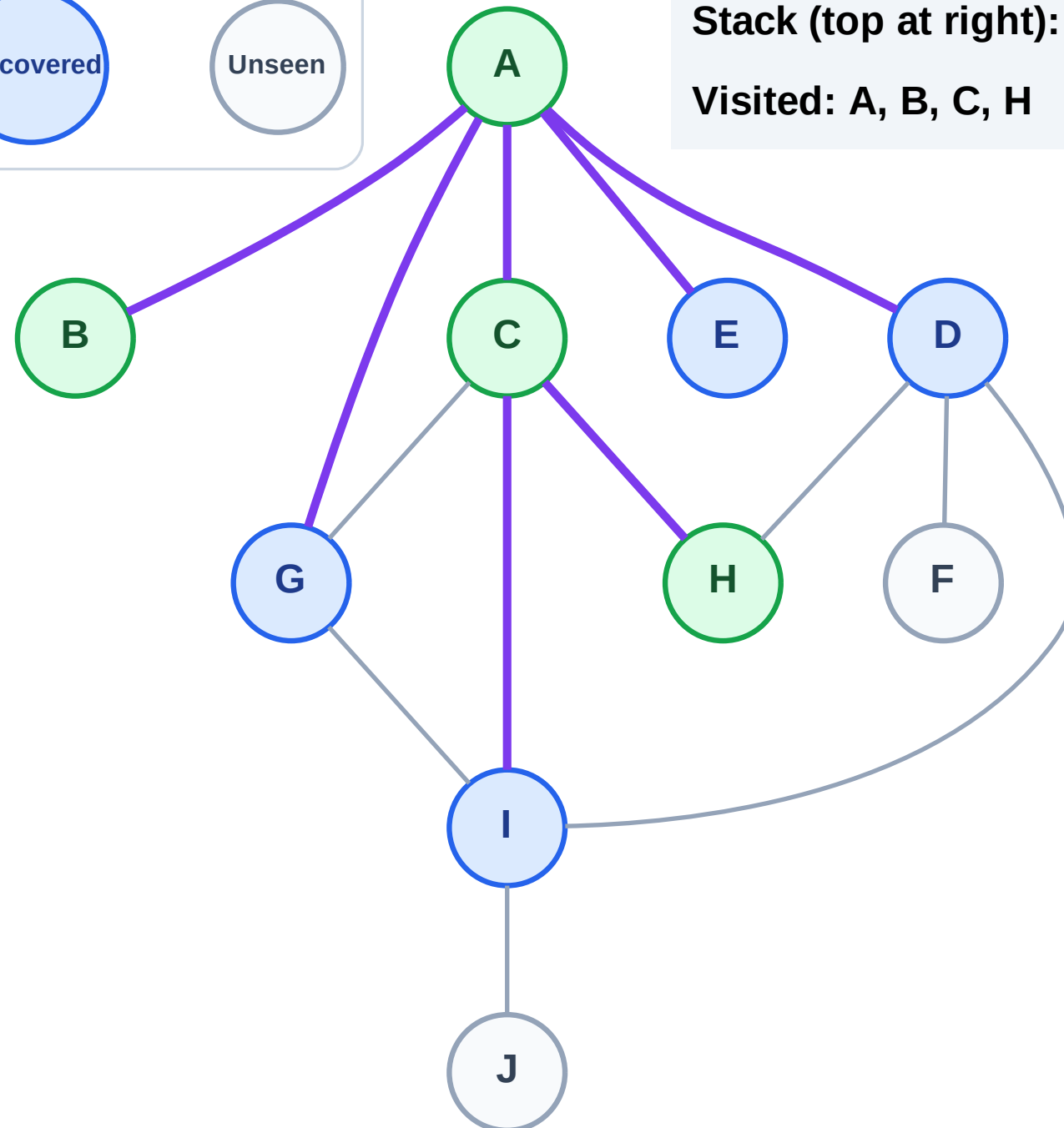
Step 9: No new neighbors from H

Legend



Stack (top at right): G | E | D | I

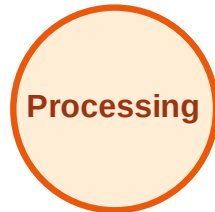
Visited: A, B, C, H



DFS Traversal

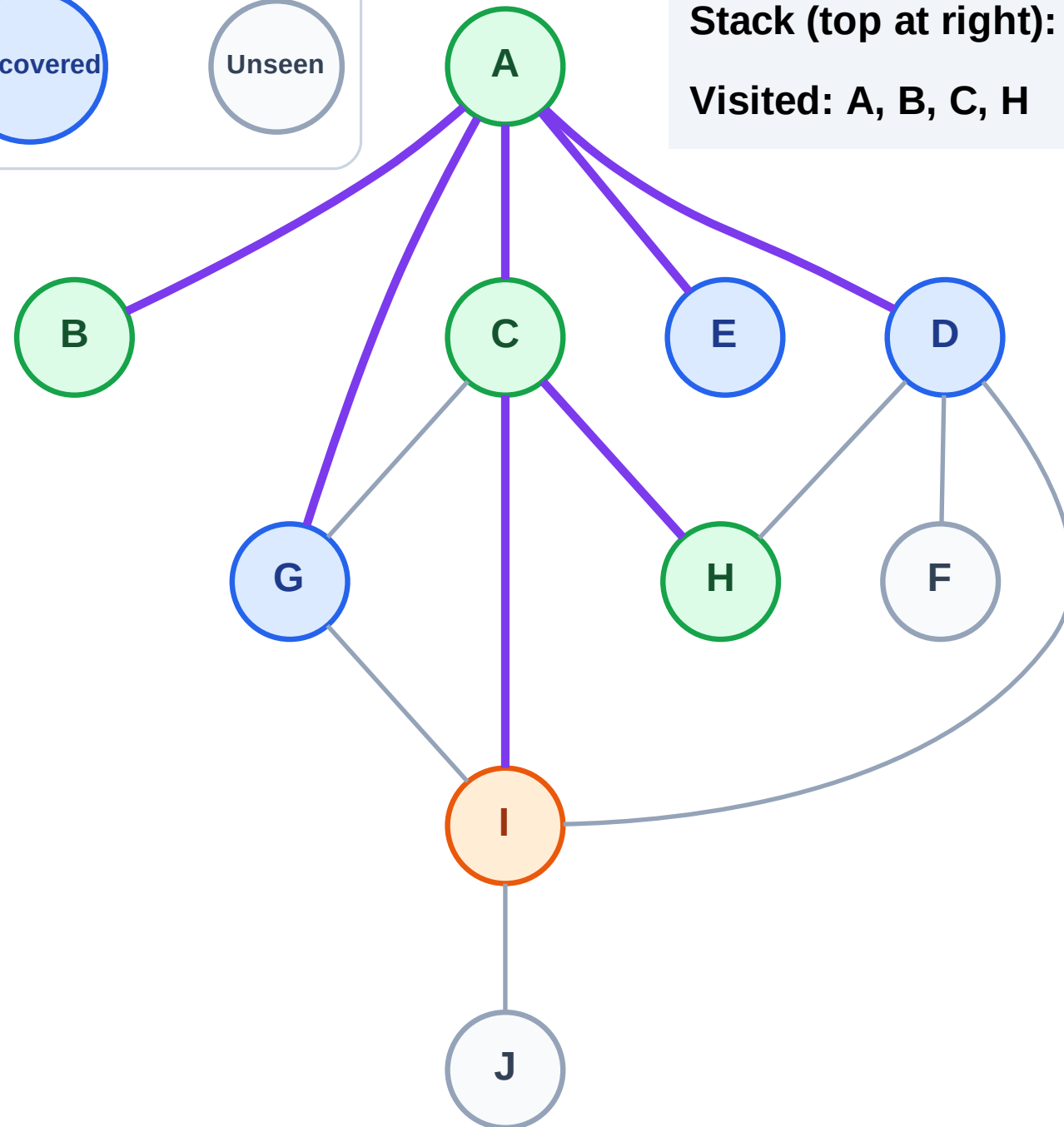
Step 10: Process node I

Legend



Stack (top at right): G | E | D

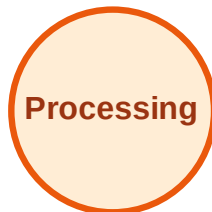
Visited: A, B, C, H



DFS Traversal

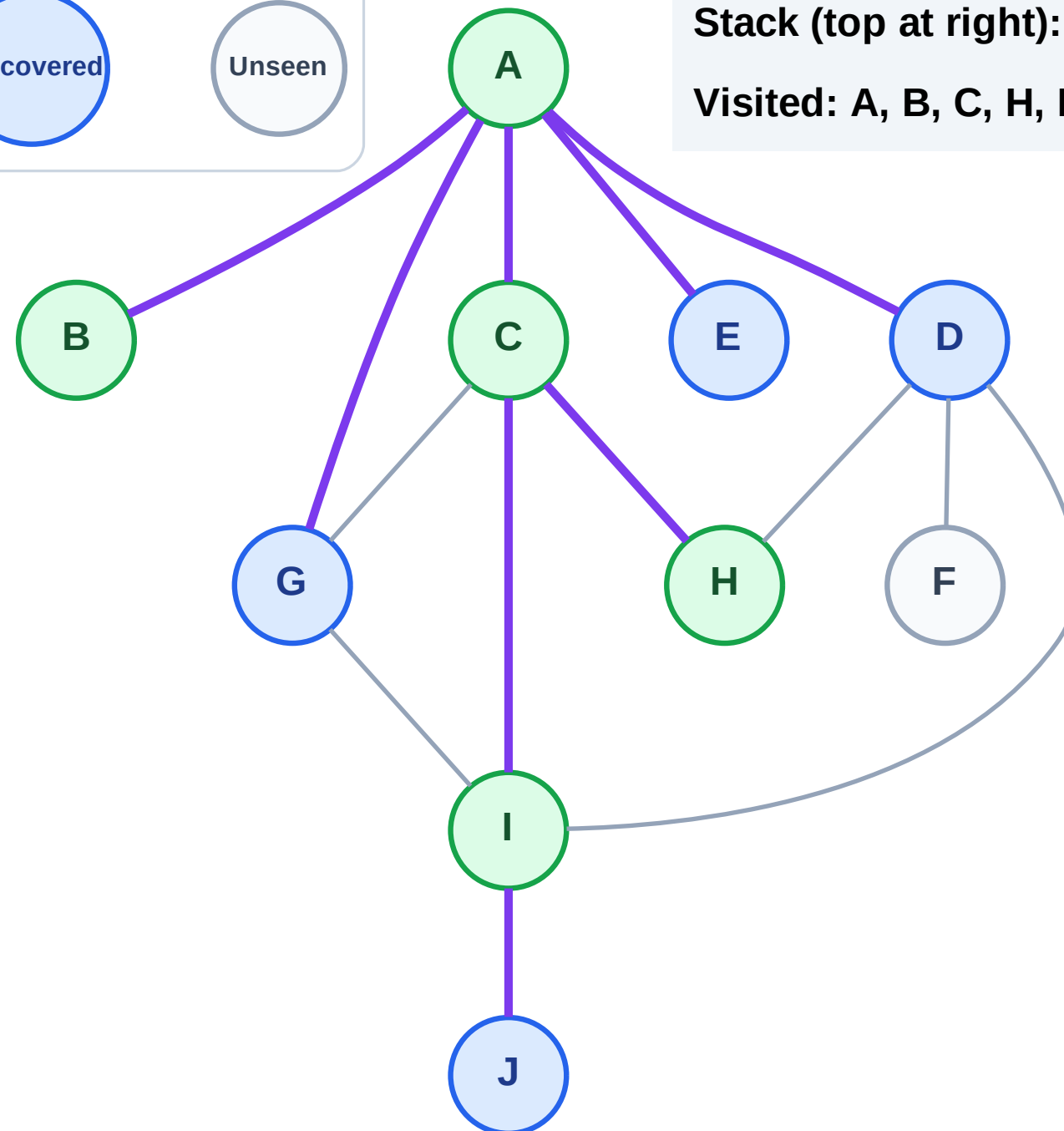
Step 11: Discover J from I

Legend



Stack (top at right): G | E | D | J

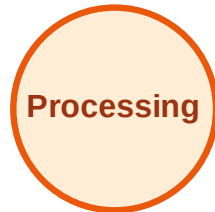
Visited: A, B, C, H, I



DFS Traversal

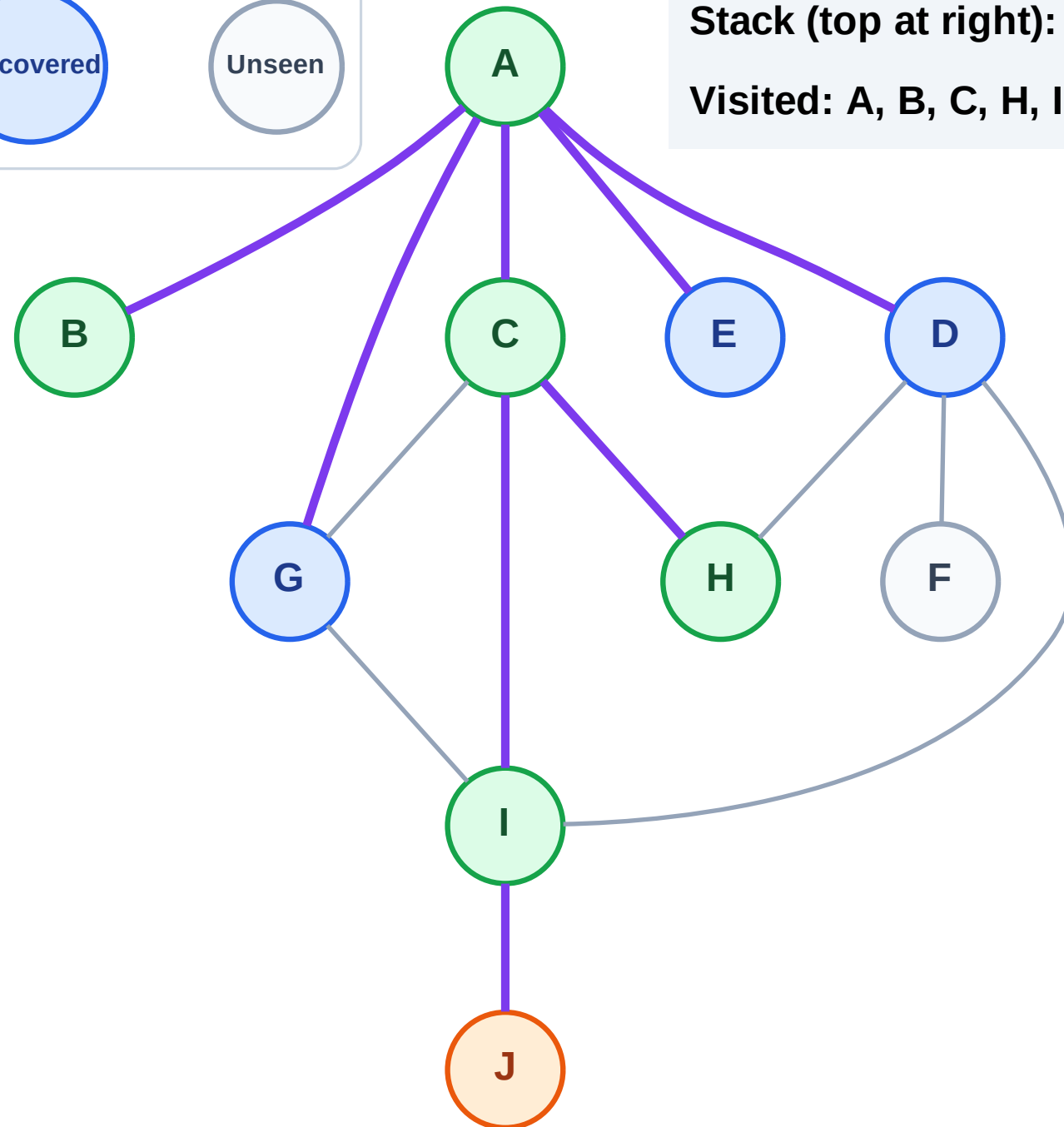
Step 12: Process node J

Legend



Stack (top at right): G | E | D

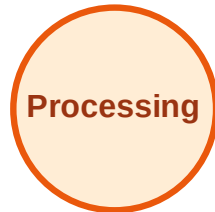
Visited: A, B, C, H, I



DFS Traversal

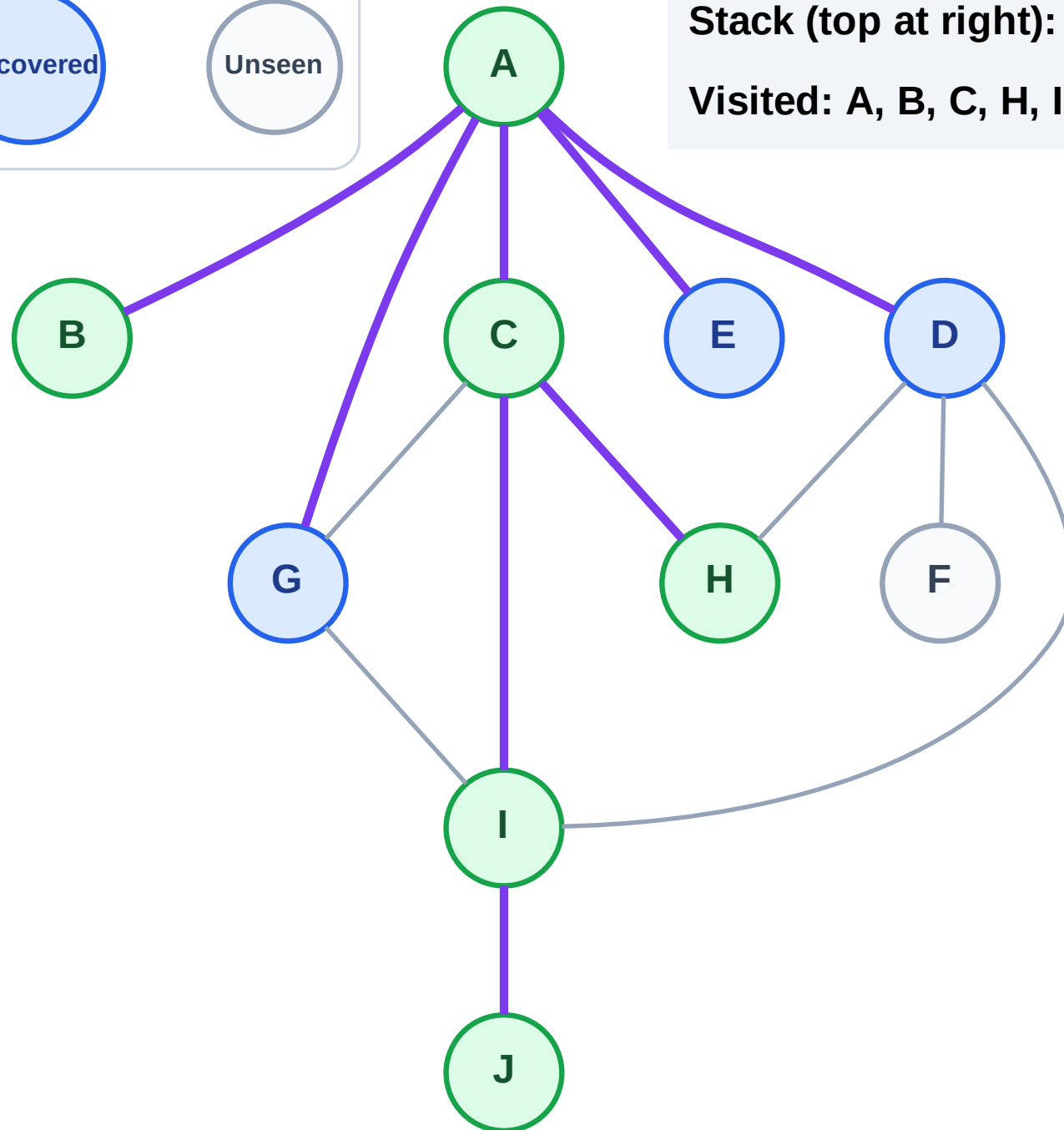
Step 13: No new neighbors from J

Legend



Stack (top at right): G | E | D

Visited: A, B, C, H, I, J



DFS Traversal

Step 14: Process node D

Legend

Complete

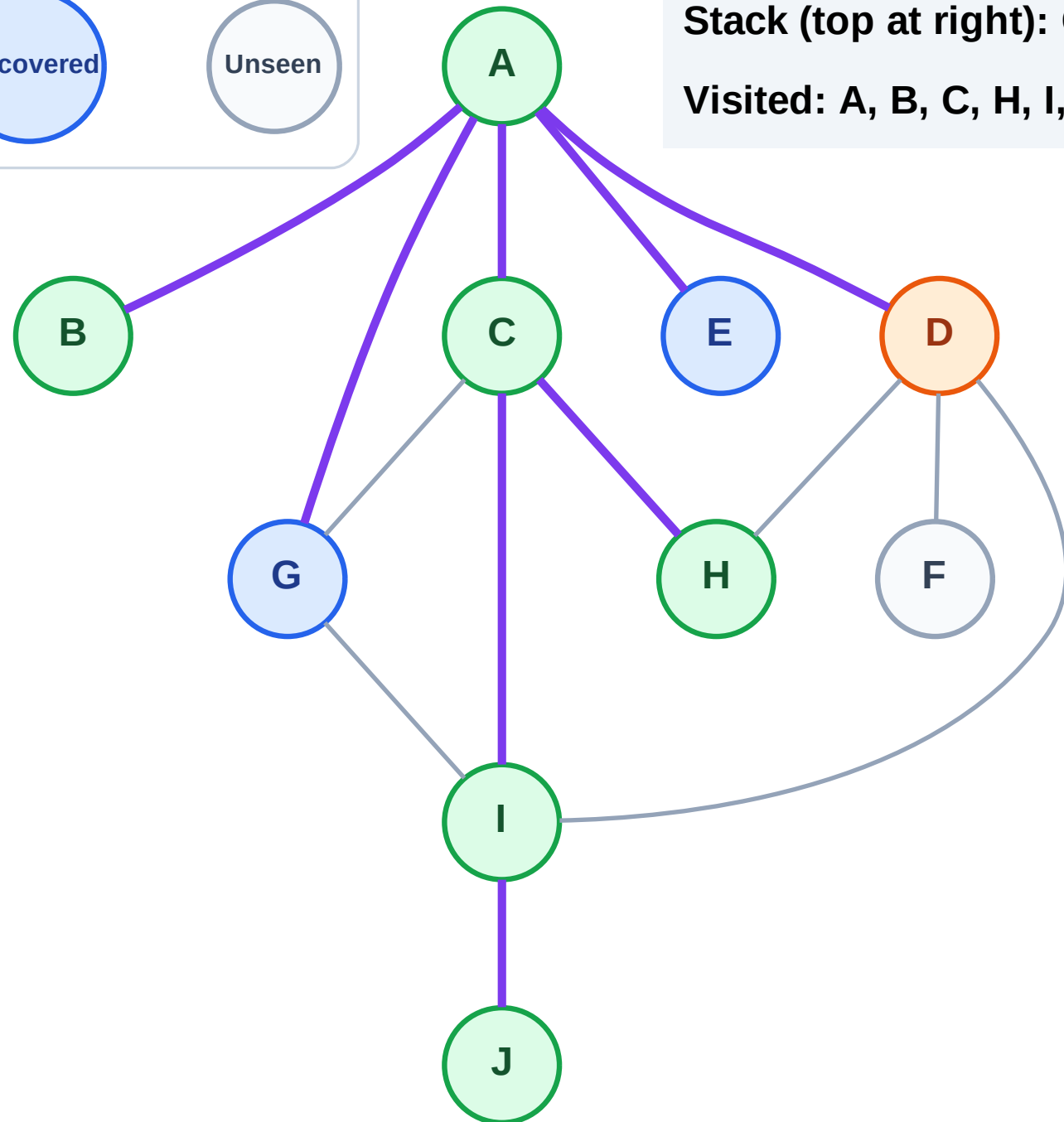
Processing

Discovered

Unseen

Stack (top at right): G | E

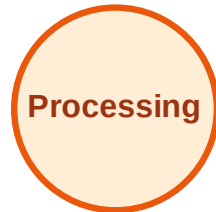
Visited: A, B, C, H, I, J



DFS Traversal

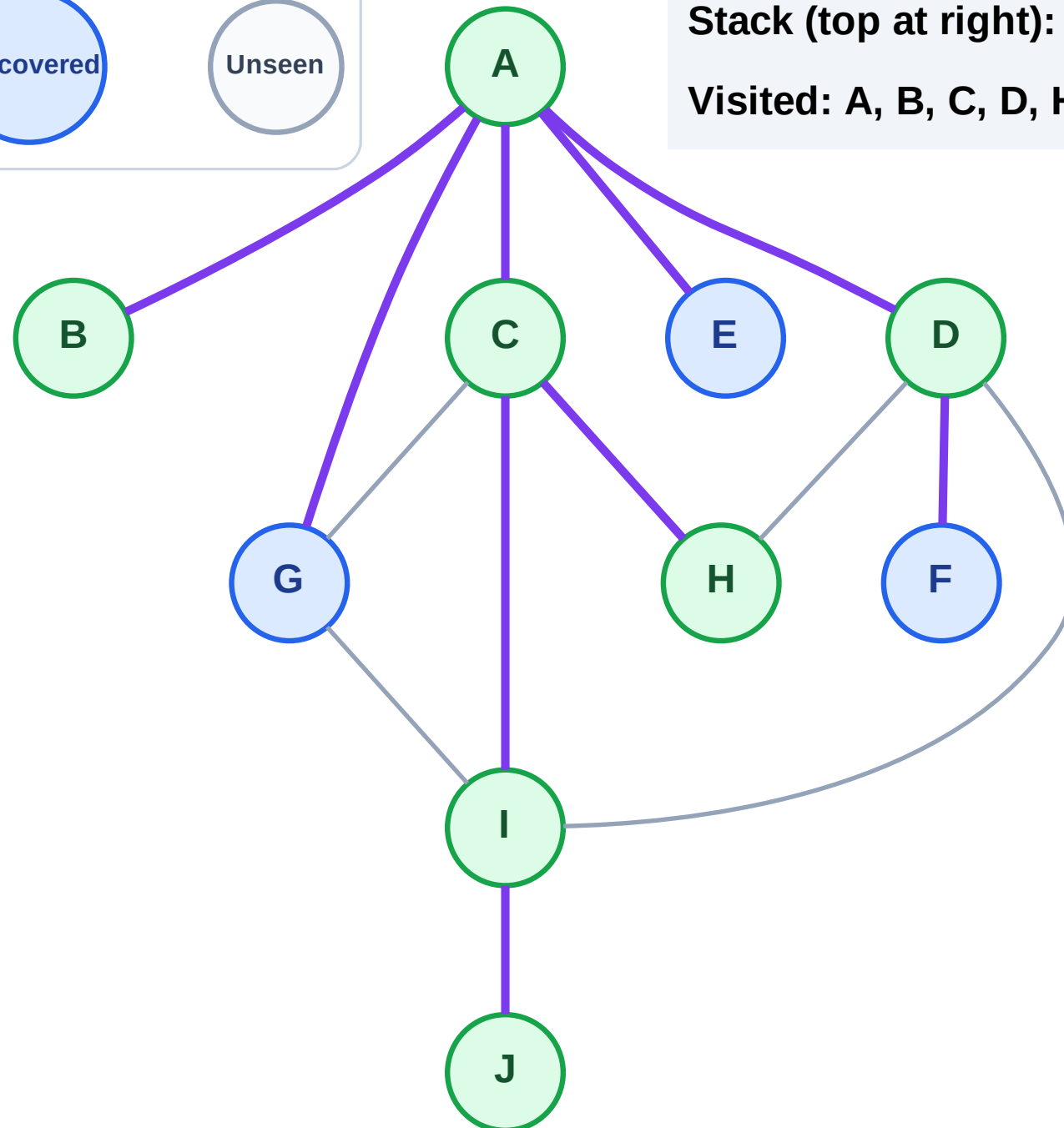
Step 15: Discover F from D

Legend



Stack (top at right): G | E | F

Visited: A, B, C, D, H, I, J



DFS Traversal

Step 16: Process node F

Legend

Complete

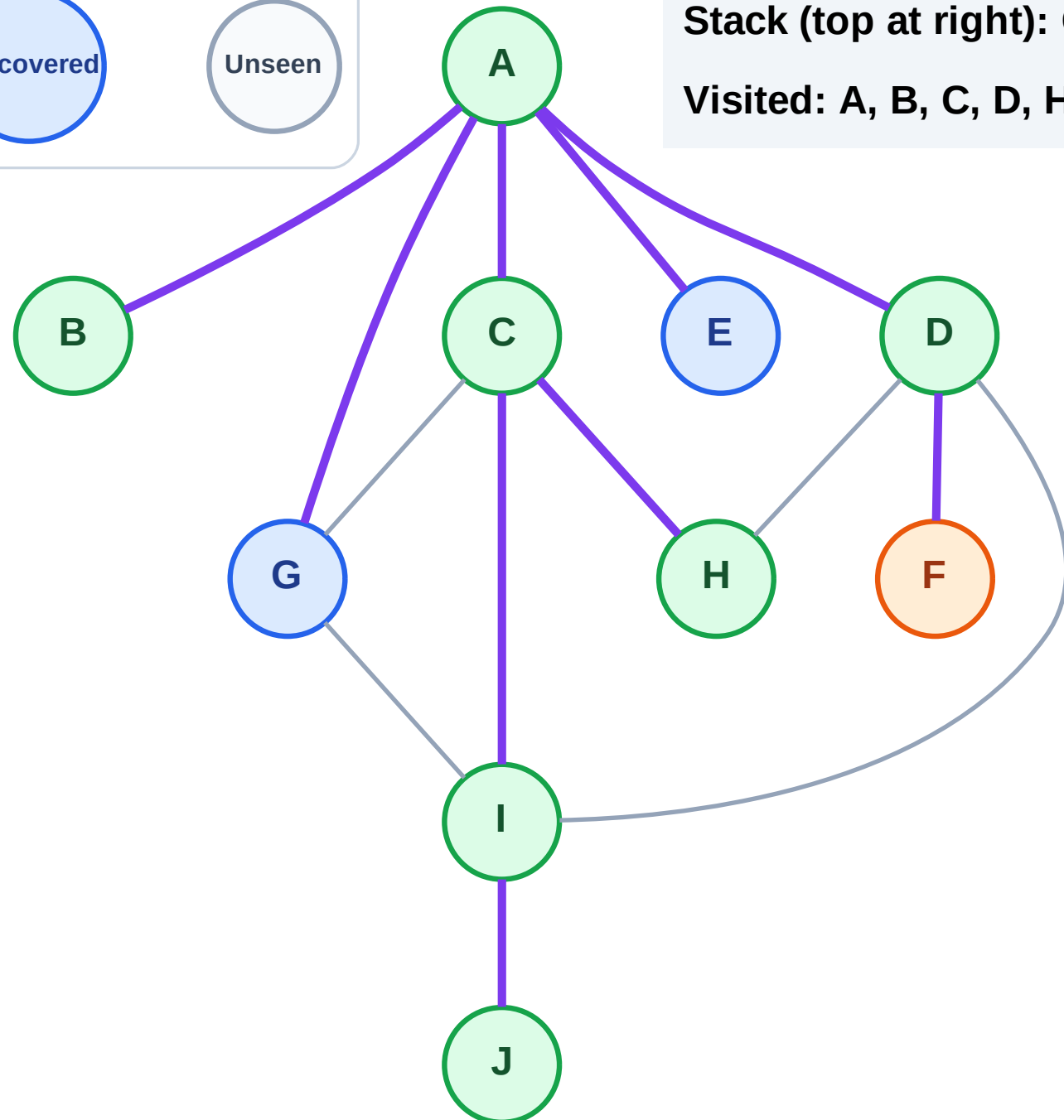
Processing

Discovered

Unseen

Stack (top at right): G | E

Visited: A, B, C, D, H, I, J



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

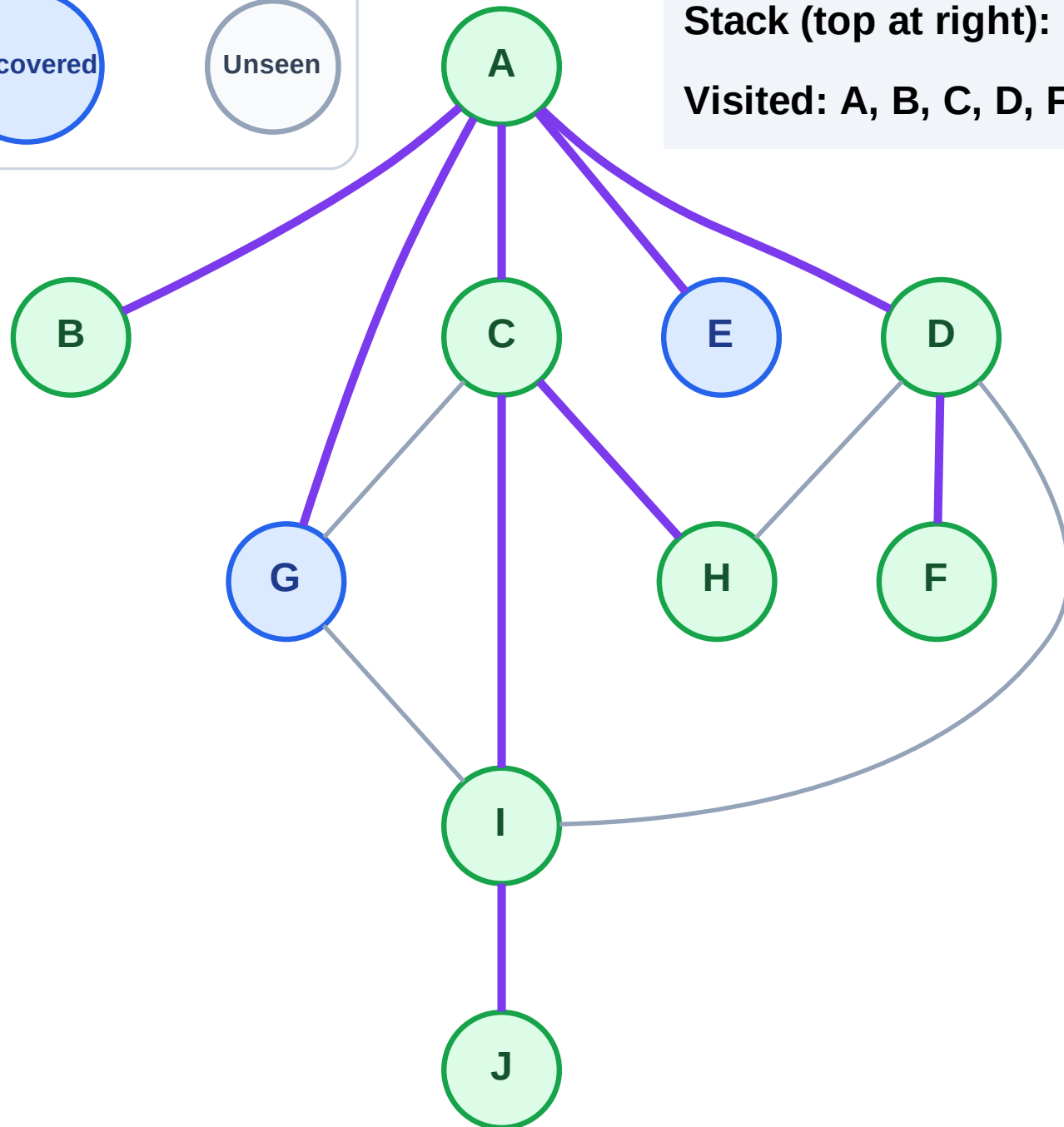
Processing

Discovered

Unseen

Stack (top at right): G | E

Visited: A, B, C, D, F, H, I, J



DFS Traversal

Step 18: Process node E

Legend

Complete

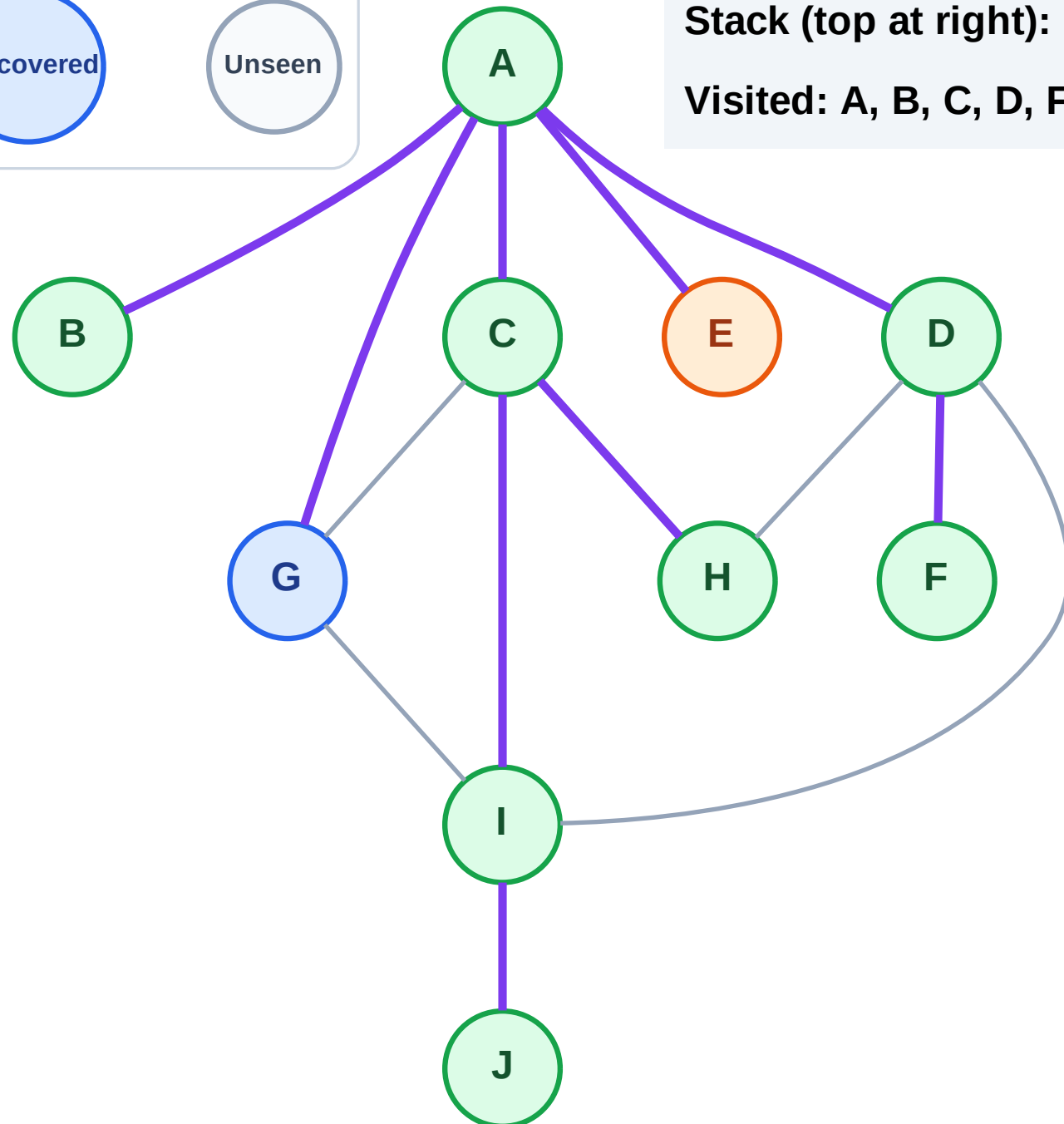
Processing

Discovered

Unseen

Stack (top at right): G

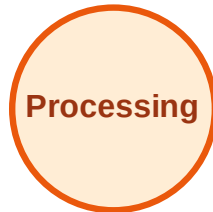
Visited: A, B, C, D, F, H, I, J



DFS Traversal

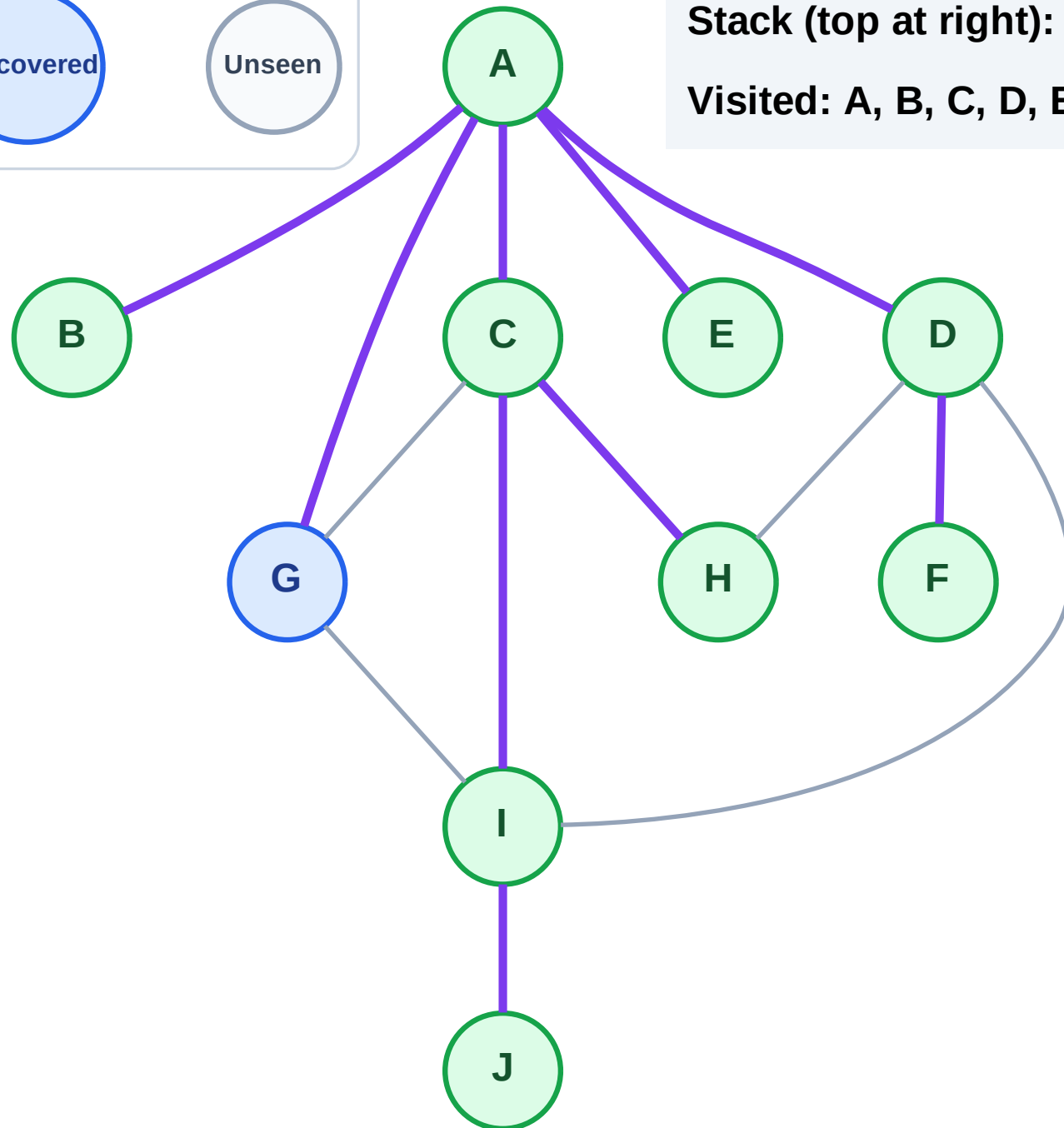
Step 19: No new neighbors from E

Legend



Stack (top at right): G

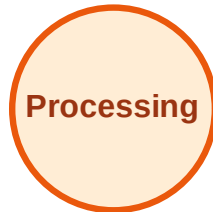
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

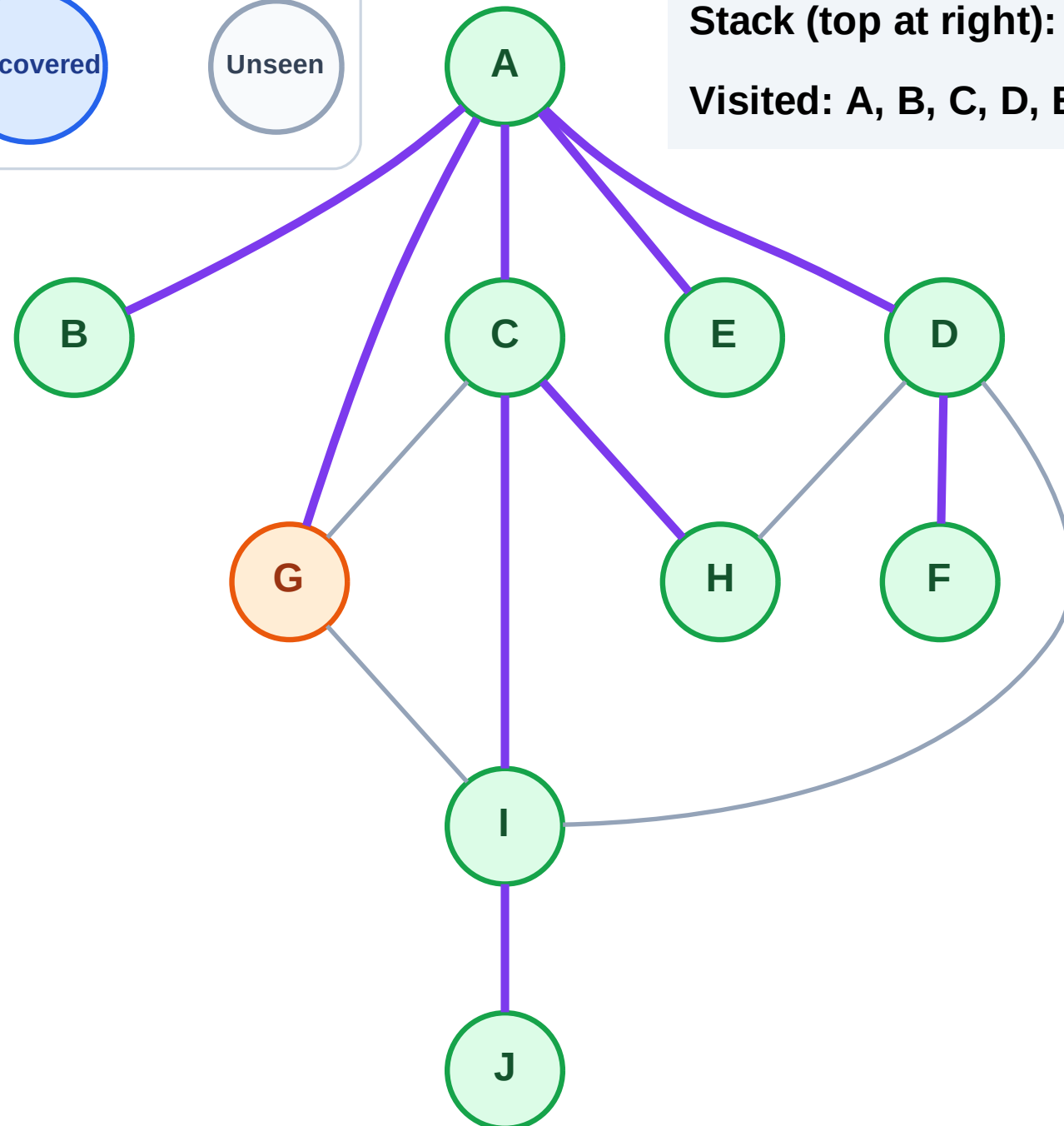
Step 20: Process node G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

